tropical: MaxSubsetSum

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1 Key results

This report covers 240 runs (5 seeds; 0 diverged) under frozen hash `ff2962d32940`.

Where to look. The headline metric is `ood_len_f1` (higher is better). Figure 1 plots it against `fishr_lambda`. Table 1 gives every cell's mean with its 95% bootstrap CI. Table 2 compares each level with the control. Table 3 gives the trend. Table 4 gives the paired contrasts. Its tests take the rank-based path: more than 20% of values within 0.01 of a bound (0, 1). Cells are compared within a seed: a seed index fixes the data and the initialization. Comparisons are marked significant at $p < 0.05$ after the stated correction.

1.1 `ood_len_f1`

Table 1: `ood_len_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	pointer	0	1	[1, 1]	3.20×10^{-5}	5	0
adaptive	pointer	0.01	1	[1, 1]	3.20×10^{-5}	5	0
adaptive	pointer	0.1	1	[1, 1]	3.20×10^{-5}	5	0
adaptive	pointer	1	1	[1, 1]	3.20×10^{-5}	5	0
adaptive	pointer	10	1	[1, 1]	3.11×10^{-5}	5	0
adaptive	pointer	100	1	[1, 1]	3.20×10^{-5}	5	0
adaptive	pointer	1000	1	[1, 1]	3.28×10^{-5}	5	0
adaptive	pointer	1e+04	1	[1, 1]	3.28×10^{-5}	5	0
adaptive	seq2seq	0	1	[1, 1]	0	5	0
adaptive	seq2seq	0.01	1	[1, 1]	0	5	0
adaptive	seq2seq	0.1	1	[1, 1]	0	5	0
adaptive	seq2seq	1	1	[1, 1]	0	5	0
adaptive	seq2seq	10	1	[1, 1]	0	5	0
adaptive	seq2seq	100	1	[1, 1]	0	5	0
adaptive	seq2seq	1000	1	[1, 1]	0	5	0
adaptive	seq2seq	1e+04	1	[1, 1]	0	5	0
tropical	pointer	0	1	[1, 1]	3.20×10^{-5}	5	0
tropical	pointer	0.01	1	[1, 1]	3.20×10^{-5}	5	0
tropical	pointer	0.1	1	[1, 1]	3.20×10^{-5}	5	0
tropical	pointer	1	1	[1, 1]	3.20×10^{-5}	5	0
tropical	pointer	10	1	[1, 1]	3.20×10^{-5}	5	0
tropical	pointer	100	1	[1, 1]	3.20×10^{-5}	5	0
tropical	pointer	1000	1	[1, 1]	3.20×10^{-5}	5	0
tropical	pointer	1e+04	1	[1, 1]	3.20×10^{-5}	5	0
tropical	seq2seq	0	1	[1, 1]	0	5	0
tropical	seq2seq	0.01	1	[1, 1]	0	5	0
tropical	seq2seq	0.1	1	[1, 1]	0	5	0
tropical	seq2seq	1	1	[1, 1]	0	5	0
tropical	seq2seq	10	1	[1, 1]	0	5	0
tropical	seq2seq	100	1	[1, 1]	0	5	0

Table 1: `ood_len_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
tropical	seq2seq	1000	1	[1, 1]	0	5	0
tropical	seq2seq	1e+04	1	[1, 1]	0	5	0
vanilla	pointer	0	1	[1, 1]	3.20×10^{-5}	5	0
vanilla	pointer	0.01	1	[1, 1]	3.20×10^{-5}	5	0
vanilla	pointer	0.1	1	[1, 1]	3.20×10^{-5}	5	0
vanilla	pointer	1	1	[1, 1]	3.20×10^{-5}	5	0
vanilla	pointer	10	1	[1, 1]	3.20×10^{-5}	5	0
vanilla	pointer	100	1	[1, 1]	3.20×10^{-5}	5	0
vanilla	pointer	1000	1	[1, 1]	3.20×10^{-5}	5	0
vanilla	pointer	1e+04	1	[1, 1]	3.20×10^{-5}	5	0
vanilla	seq2seq	0	1	[1, 1]	0	5	0
vanilla	seq2seq	0.01	1	[1, 1]	0	5	0
vanilla	seq2seq	0.1	1	[1, 1]	0	5	0
vanilla	seq2seq	1	1	[1, 1]	0	5	0
vanilla	seq2seq	10	1	[1, 1]	0	5	0
vanilla	seq2seq	100	1	[1, 1]	0	5	0
vanilla	seq2seq	1000	1	[1, 1]	0	5	0
vanilla	seq2seq	1e+04	1	[1, 1]	0	5	0

Table 2: `ood_len_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. rank: Wilcoxon signed-rank by seed, Holm.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	0	[0, 0]	–
adaptive	pointer	0.1	0	[0, 0]	–
adaptive	pointer	1	0	[0, 0]	–
adaptive	pointer	10	0	$[-7.16 \times 10^{-6}, 0]$	1.0000
adaptive	pointer	100	0	[0, 0]	–
adaptive	pointer	1000	0	$[-2.51 \times 10^{-5}, 0]$	1.0000
adaptive	pointer	1e+04	0	$[-2.51 \times 10^{-5}, 0]$	1.0000
adaptive	seq2seq	0.01	0	[0, 0]	–
adaptive	seq2seq	0.1	0	[0, 0]	–
adaptive	seq2seq	1	0	[0, 0]	–
adaptive	seq2seq	10	0	[0, 0]	–
adaptive	seq2seq	100	0	[0, 0]	–
adaptive	seq2seq	1000	0	[0, 0]	–
adaptive	seq2seq	1e+04	0	[0, 0]	–
tropical	pointer	0.01	0	[0, 0]	–
tropical	pointer	0.1	0	[0, 0]	–
tropical	pointer	1	0	[0, 0]	–
tropical	pointer	10	0	[0, 0]	–
tropical	pointer	100	0	[0, 0]	–
tropical	pointer	1000	0	[0, 0]	–
tropical	pointer	1e+04	0	[0, 0]	–
tropical	seq2seq	0.01	0	[0, 0]	–
tropical	seq2seq	0.1	0	[0, 0]	–
tropical	seq2seq	1	0	[0, 0]	–
tropical	seq2seq	10	0	[0, 0]	–
tropical	seq2seq	100	0	[0, 0]	–
tropical	seq2seq	1000	0	[0, 0]	–
tropical	seq2seq	1e+04	0	[0, 0]	–
vanilla	pointer	0.01	0	[0, 0]	–
vanilla	pointer	0.1	0	[0, 0]	–

Table 2: `ood_len_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. rank: Wilcoxon signed-rank by seed, Holm.

model_type	head	fishr_lambda	difference	95% CI	p
vanilla	pointer	1	0	[0, 0]	–
vanilla	pointer	10	0	[0, 0]	–
vanilla	pointer	100	0	[0, 0]	–
vanilla	pointer	1000	0	[0, 0]	–
vanilla	pointer	1e+04	0	[0, 0]	–
vanilla	seq2seq	0.01	0	[0, 0]	–
vanilla	seq2seq	0.1	0	[0, 0]	–
vanilla	seq2seq	1	0	[0, 0]	–
vanilla	seq2seq	10	0	[0, 0]	–
vanilla	seq2seq	100	0	[0, 0]	–
vanilla	seq2seq	1000	0	[0, 0]	–
vanilla	seq2seq	1e+04	0	[0, 0]	–

Table 3: `ood_len_f1` against log10 `fishr_lambda`: rank: Spearman rho, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

model_type	head	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	-0.1294	[-0.527, 0]	0.4587	-0.08	0.4603
adaptive	seq2seq	–	[–, –]	–	–	–
tropical	pointer	0	[0, 0]	1.0000	0	1.0000
tropical	seq2seq	–	[–, –]	–	–	–
vanilla	pointer	0	[0, 0]	1.0000	0	1.0000
vanilla	seq2seq	–	[–, –]	–	–	–

Table 4: `ood_len_f1`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	0	[–, –]	–
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0	[–, –]	–
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0	[–, –]	–
model_type: tropical - vanilla head=pointer fishr_lambda=1	0	[–, –]	–
model_type: tropical - vanilla head=pointer fishr_lambda=10	0	[–, –]	–
model_type: tropical - vanilla head=pointer fishr_lambda=100	0	[–, –]	–
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0	[–, –]	–
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0	[–, –]	–
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0	[–, –]	–
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0	[–, –]	–
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0	[–, –]	–
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0	[–, –]	–

Table 4: `ood_len_f1`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0	[–, –]	–
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0	[–, –]	–
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	0	[–, –]	–
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0	[–, –]	–
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0	[–, –]	–
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	0	[–, –]	–
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	0	[–, –]	–
model_type: adaptive - vanilla head=pointer fishr_lambda=1	0	[–, –]	–
model_type: adaptive - vanilla head=pointer fishr_lambda=10	-2.387×10^{-6}	$[-9.01 \times 10^{-6}, 4.24 \times 10^{-6}]$	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=100	0	[–, –]	–
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	-8.354×10^{-6}	$[-3.15 \times 10^{-5}, 1.48 \times 10^{-5}]$	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	-8.354×10^{-6}	$[-3.15 \times 10^{-5}, 1.48 \times 10^{-5}]$	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	0	[–, –]	–
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	-1.432×10^{-5}	$[-5.41 \times 10^{-5}, 2.54 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	-1.432×10^{-5}	$[-5.41 \times 10^{-5}, 2.54 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	-1.432×10^{-5}	$[-5.41 \times 10^{-5}, 2.54 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	-1.432×10^{-5}	$[-5.41 \times 10^{-5}, 2.54 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	-1.671×10^{-5}	$[-5.54 \times 10^{-5}, 2.19 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	-1.432×10^{-5}	$[-5.41 \times 10^{-5}, 2.54 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	-2.268×10^{-5}	$[-6.34 \times 10^{-5}, 1.80 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	-2.268×10^{-5}	$[-6.34 \times 10^{-5}, 1.80 \times 10^{-5}]$	1.0000

Table 4: `ood_len_f1`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
head: pointer - seq2seq model_type=tropical fishr_lambda=0	-1.432×10^{-5}	$[-5.41 \times 10^{-5}, 2.54 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	-1.432×10^{-5}	$[-5.41 \times 10^{-5}, 2.54 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	-1.432×10^{-5}	$[-5.41 \times 10^{-5}, 2.54 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1	-1.432×10^{-5}	$[-5.41 \times 10^{-5}, 2.54 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=10	-1.432×10^{-5}	$[-5.41 \times 10^{-5}, 2.54 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=100	-1.432×10^{-5}	$[-5.41 \times 10^{-5}, 2.54 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	-1.432×10^{-5}	$[-5.41 \times 10^{-5}, 2.54 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	-1.432×10^{-5}	$[-5.41 \times 10^{-5}, 2.54 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	-1.432×10^{-5}	$[-5.41 \times 10^{-5}, 2.54 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	-1.432×10^{-5}	$[-5.41 \times 10^{-5}, 2.54 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	-1.432×10^{-5}	$[-5.41 \times 10^{-5}, 2.54 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	-1.432×10^{-5}	$[-5.41 \times 10^{-5}, 2.54 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	-1.432×10^{-5}	$[-5.41 \times 10^{-5}, 2.54 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	-1.432×10^{-5}	$[-5.41 \times 10^{-5}, 2.54 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	-1.432×10^{-5}	$[-5.41 \times 10^{-5}, 2.54 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	-1.432×10^{-5}	$[-5.41 \times 10^{-5}, 2.54 \times 10^{-5}]$	1.0000

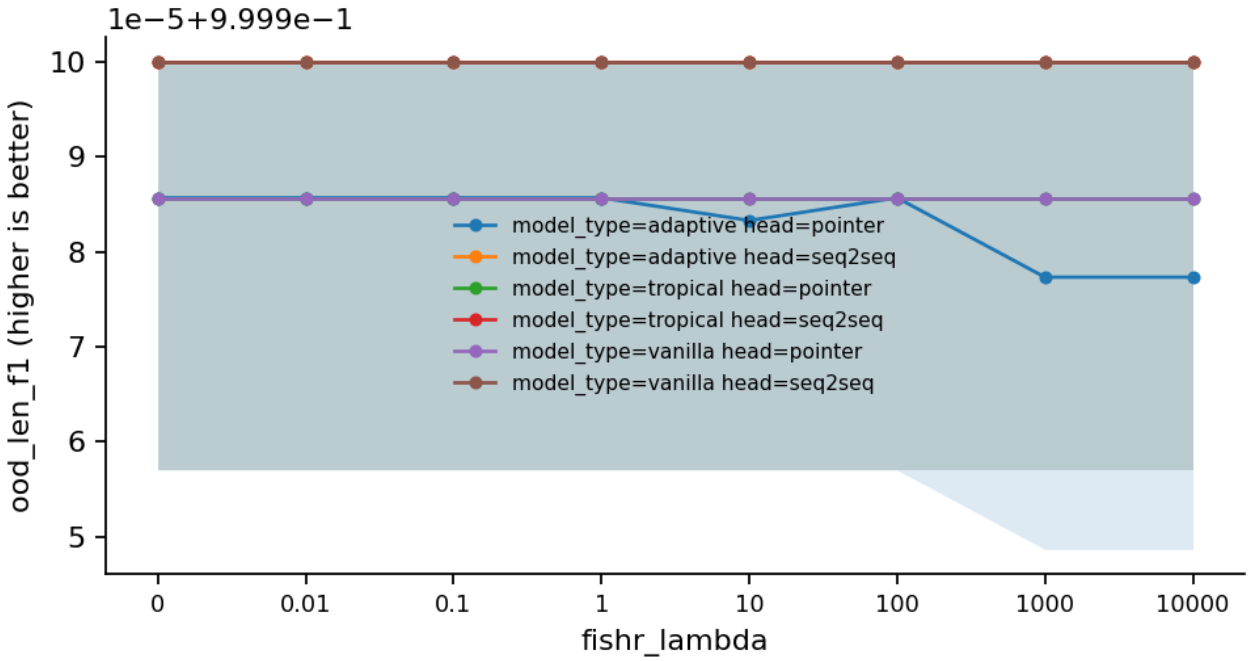


Figure 1: `ood_len_f1` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

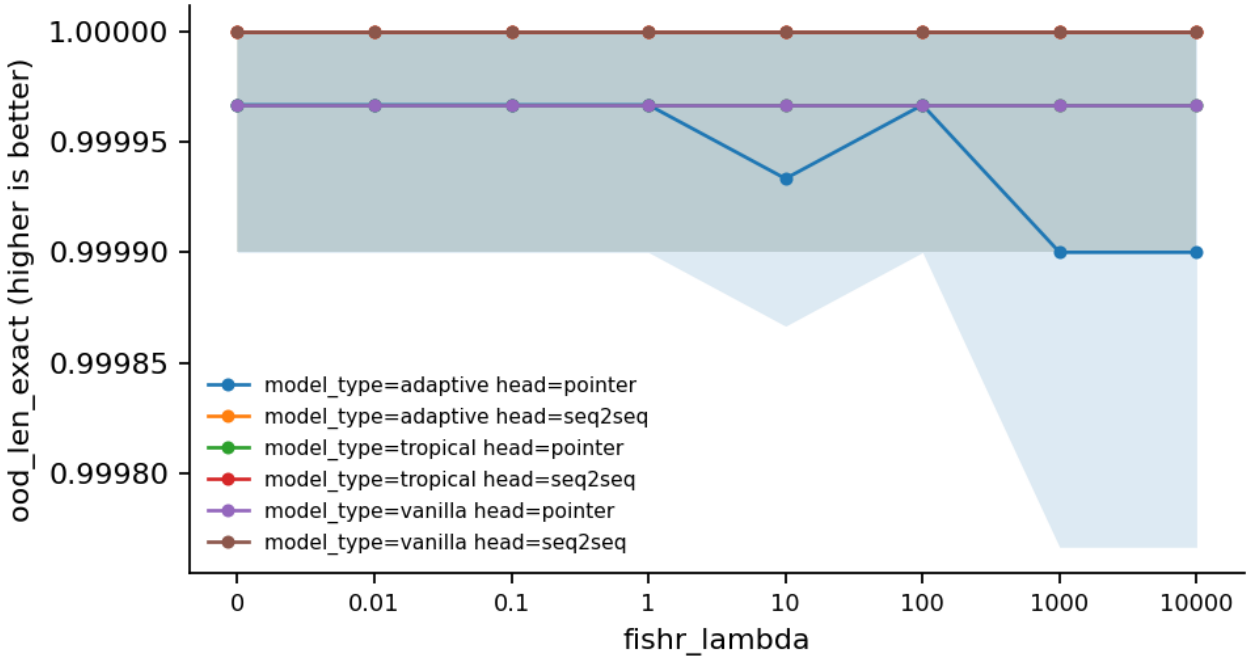


Figure 2: `ood_len_exact` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

A Other metrics

A.1 `ood_len_exact`

Table 5: `ood_len_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	1	[1, 1]	7.45×10^{-5}	5	0
adaptive	pointer	0.01	1	[1, 1]	7.45×10^{-5}	5	0
adaptive	pointer	0.1	1	[1, 1]	7.45×10^{-5}	5	0
adaptive	pointer	1	1	[1, 1]	7.45×10^{-5}	5	0
adaptive	pointer	10	0.9999	[1, 1]	9.13×10^{-5}	5	0
adaptive	pointer	100	1	[1, 1]	7.45×10^{-5}	5	0
adaptive	pointer	1000	0.9999	[1, 1]	1.49×10^{-4}	5	0
adaptive	pointer	1e+04	0.9999	[1, 1]	1.49×10^{-4}	5	0
adaptive	seq2seq	0	1	[1, 1]	0	5	0
adaptive	seq2seq	0.01	1	[1, 1]	0	5	0
adaptive	seq2seq	0.1	1	[1, 1]	0	5	0
adaptive	seq2seq	1	1	[1, 1]	0	5	0
adaptive	seq2seq	10	1	[1, 1]	0	5	0
adaptive	seq2seq	100	1	[1, 1]	0	5	0
adaptive	seq2seq	1000	1	[1, 1]	0	5	0
adaptive	seq2seq	1e+04	1	[1, 1]	0	5	0
tropical	pointer	0	1	[1, 1]	7.45×10^{-5}	5	0
tropical	pointer	0.01	1	[1, 1]	7.45×10^{-5}	5	0
tropical	pointer	0.1	1	[1, 1]	7.45×10^{-5}	5	0
tropical	pointer	1	1	[1, 1]	7.45×10^{-5}	5	0
tropical	pointer	10	1	[1, 1]	7.45×10^{-5}	5	0
tropical	pointer	100	1	[1, 1]	7.45×10^{-5}	5	0

Table 5: `ood_len_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
tropical	pointer	1000	1	[1, 1]	7.45×10^{-5}	5	0
tropical	pointer	1e+04	1	[1, 1]	7.45×10^{-5}	5	0
tropical	seq2seq	0	1	[1, 1]	0	5	0
tropical	seq2seq	0.01	1	[1, 1]	0	5	0
tropical	seq2seq	0.1	1	[1, 1]	0	5	0
tropical	seq2seq	1	1	[1, 1]	0	5	0
tropical	seq2seq	10	1	[1, 1]	0	5	0
tropical	seq2seq	100	1	[1, 1]	0	5	0
tropical	seq2seq	1000	1	[1, 1]	0	5	0
tropical	seq2seq	1e+04	1	[1, 1]	0	5	0
vanilla	pointer	0	1	[1, 1]	7.45×10^{-5}	5	0
vanilla	pointer	0.01	1	[1, 1]	7.45×10^{-5}	5	0
vanilla	pointer	0.1	1	[1, 1]	7.45×10^{-5}	5	0
vanilla	pointer	1	1	[1, 1]	7.45×10^{-5}	5	0
vanilla	pointer	10	1	[1, 1]	7.45×10^{-5}	5	0
vanilla	pointer	100	1	[1, 1]	7.45×10^{-5}	5	0
vanilla	pointer	1000	1	[1, 1]	7.45×10^{-5}	5	0
vanilla	pointer	1e+04	1	[1, 1]	7.45×10^{-5}	5	0
vanilla	seq2seq	0	1	[1, 1]	0	5	0
vanilla	seq2seq	0.01	1	[1, 1]	0	5	0
vanilla	seq2seq	0.1	1	[1, 1]	0	5	0
vanilla	seq2seq	1	1	[1, 1]	0	5	0
vanilla	seq2seq	10	1	[1, 1]	0	5	0
vanilla	seq2seq	100	1	[1, 1]	0	5	0
vanilla	seq2seq	1000	1	[1, 1]	0	5	0
vanilla	seq2seq	1e+04	1	[1, 1]	0	5	0

Table 6: `ood_len_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. rank: Wilcoxon signed-rank by seed, Holm.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	0	[0, 0]	–
adaptive	pointer	0.1	0	[0, 0]	–
adaptive	pointer	1	0	[0, 0]	–
adaptive	pointer	10	0	$[-1.00 \times 10^{-4}, 0]$	1.0000
adaptive	pointer	100	0	[0, 0]	–
adaptive	pointer	1000	0	$[-2.00 \times 10^{-4}, 0]$	1.0000
adaptive	pointer	1e+04	0	$[-2.00 \times 10^{-4}, 0]$	1.0000
adaptive	seq2seq	0.01	0	[0, 0]	–
adaptive	seq2seq	0.1	0	[0, 0]	–
adaptive	seq2seq	1	0	[0, 0]	–
adaptive	seq2seq	10	0	[0, 0]	–
adaptive	seq2seq	100	0	[0, 0]	–
adaptive	seq2seq	1000	0	[0, 0]	–
adaptive	seq2seq	1e+04	0	[0, 0]	–
tropical	pointer	0.01	0	[0, 0]	–
tropical	pointer	0.1	0	[0, 0]	–
tropical	pointer	1	0	[0, 0]	–
tropical	pointer	10	0	[0, 0]	–
tropical	pointer	100	0	[0, 0]	–
tropical	pointer	1000	0	[0, 0]	–
tropical	pointer	1e+04	0	[0, 0]	–
tropical	seq2seq	0.01	0	[0, 0]	–

Table 6: `ood_len_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. rank: Wilcoxon signed-rank by seed, Holm.

`model_type`	`head`	`fishr_lambda`	difference	95% CI	p
tropical	seq2seq	0.1	0	[0, 0]	–
tropical	seq2seq	1	0	[0, 0]	–
tropical	seq2seq	10	0	[0, 0]	–
tropical	seq2seq	100	0	[0, 0]	–
tropical	seq2seq	1000	0	[0, 0]	–
tropical	seq2seq	1e+04	0	[0, 0]	–
vanilla	pointer	0.01	0	[0, 0]	–
vanilla	pointer	0.1	0	[0, 0]	–
vanilla	pointer	1	0	[0, 0]	–
vanilla	pointer	10	0	[0, 0]	–
vanilla	pointer	100	0	[0, 0]	–
vanilla	pointer	1000	0	[0, 0]	–
vanilla	pointer	1e+04	0	[0, 0]	–
vanilla	seq2seq	0.01	0	[0, 0]	–
vanilla	seq2seq	0.1	0	[0, 0]	–
vanilla	seq2seq	1	0	[0, 0]	–
vanilla	seq2seq	10	0	[0, 0]	–
vanilla	seq2seq	100	0	[0, 0]	–
vanilla	seq2seq	1000	0	[0, 0]	–
vanilla	seq2seq	1e+04	0	[0, 0]	–

Table 7: `ood_len_exact` against log10 `fishr_lambda`: rank: Spearman rho, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	-0.1925	[-0.53, 0]	0.2679	-0.118	0.2743
adaptive	seq2seq	–	[–, –]	–	–	–
tropical	pointer	0	[0, 0]	1.0000	0	1.0000
tropical	seq2seq	–	[–, –]	–	–	–
vanilla	pointer	0	[0, 0]	1.0000	0	1.0000
vanilla	seq2seq	–	[–, –]	–	–	–

Table 8: `ood_len_exact`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	0	[–, –]	–
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0	[–, –]	–
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0	[–, –]	–
model_type: tropical - vanilla head=pointer fishr_lambda=1	0	[–, –]	–
model_type: tropical - vanilla head=pointer fishr_lambda=10	0	[–, –]	–
model_type: tropical - vanilla head=pointer fishr_lambda=100	0	[–, –]	–
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0	[–, –]	–

Table 8: `ood_len_exact`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0	[–, –]	–
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0	[–, –]	–
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0	[–, –]	–
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0	[–, –]	–
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0	[–, –]	–
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0	[–, –]	–
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0	[–, –]	–
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	0	[–, –]	–
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0	[–, –]	–
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0	[–, –]	–
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	0	[–, –]	–
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	0	[–, –]	–
model_type: adaptive - vanilla head=pointer fishr_lambda=1	0	[–, –]	–
model_type: adaptive - vanilla head=pointer fishr_lambda=10	-3.333×10^{-5}	$[-1.26 \times 10^{-4}, 5.92 \times 10^{-5}]$	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=100	0	[–, –]	–
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	-6.667×10^{-5}	$[-2.52 \times 10^{-4}, 1.18 \times 10^{-4}]$	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	-6.667×10^{-5}	$[-2.52 \times 10^{-4}, 1.18 \times 10^{-4}]$	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	0	[–, –]	–
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	-3.333×10^{-5}	$[-1.26 \times 10^{-4}, 5.92 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	-3.333×10^{-5}	$[-1.26 \times 10^{-4}, 5.92 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	-3.333×10^{-5}	$[-1.26 \times 10^{-4}, 5.92 \times 10^{-5}]$	1.0000

Table 8: `ood_len_exact`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	-3.333×10^{-5}	$[-1.26 \times 10^{-4}, 5.92 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	-6.667×10^{-5}	$[-1.80 \times 10^{-4}, 4.67 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	-3.333×10^{-5}	$[-1.26 \times 10^{-4}, 5.92 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	-1.000×10^{-4}	$[-2.85 \times 10^{-4}, 8.51 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	-1.000×10^{-4}	$[-2.85 \times 10^{-4}, 8.51 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0	-3.333×10^{-5}	$[-1.26 \times 10^{-4}, 5.92 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	-3.333×10^{-5}	$[-1.26 \times 10^{-4}, 5.92 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	-3.333×10^{-5}	$[-1.26 \times 10^{-4}, 5.92 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1	-3.333×10^{-5}	$[-1.26 \times 10^{-4}, 5.92 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=10	-3.333×10^{-5}	$[-1.26 \times 10^{-4}, 5.92 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=100	-3.333×10^{-5}	$[-1.26 \times 10^{-4}, 5.92 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	-3.333×10^{-5}	$[-1.26 \times 10^{-4}, 5.92 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	-3.333×10^{-5}	$[-1.26 \times 10^{-4}, 5.92 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	-3.333×10^{-5}	$[-1.26 \times 10^{-4}, 5.92 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	-3.333×10^{-5}	$[-1.26 \times 10^{-4}, 5.92 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	-3.333×10^{-5}	$[-1.26 \times 10^{-4}, 5.92 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	-3.333×10^{-5}	$[-1.26 \times 10^{-4}, 5.92 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	-3.333×10^{-5}	$[-1.26 \times 10^{-4}, 5.92 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	-3.333×10^{-5}	$[-1.26 \times 10^{-4}, 5.92 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	-3.333×10^{-5}	$[-1.26 \times 10^{-4}, 5.92 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	-3.333×10^{-5}	$[-1.26 \times 10^{-4}, 5.92 \times 10^{-5}]$	1.0000

A.2 `ood_val_f1`

Table 9: `ood_val_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.9824	[0.981, 0.983]	0.00126	5	0
adaptive	pointer	0.01	0.9811	[0.979, 0.983]	0.00238	5	0
adaptive	pointer	0.1	0.9811	[0.979, 0.983]	0.00236	5	0
adaptive	pointer	1	0.9811	[0.979, 0.983]	0.00239	5	0
adaptive	pointer	10	0.9812	[0.979, 0.983]	0.00227	5	0
adaptive	pointer	100	0.9811	[0.979, 0.983]	0.00236	5	0

Table 9: `ood_val_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	pointer	1000	0.9807	[0.978, 0.983]	0.00297	5	0
adaptive	pointer	1e+04	0.9807	[0.978, 0.983]	0.00298	5	0
adaptive	seq2seq	0	1	[1, 1]	0	5	0
adaptive	seq2seq	0.01	1	[1, 1]	0	5	0
adaptive	seq2seq	0.1	1	[1, 1]	0	5	0
adaptive	seq2seq	1	1	[1, 1]	0	5	0
adaptive	seq2seq	10	1	[1, 1]	0	5	0
adaptive	seq2seq	100	1	[1, 1]	0	5	0
adaptive	seq2seq	1000	1	[1, 1]	0	5	0
adaptive	seq2seq	1e+04	1	[1, 1]	0	5	0
tropical	pointer	0	0.9825	[0.981, 0.983]	0.00124	5	0
tropical	pointer	0.01	0.9825	[0.981, 0.983]	0.00124	5	0
tropical	pointer	0.1	0.9825	[0.981, 0.983]	0.00124	5	0
tropical	pointer	1	0.9825	[0.981, 0.983]	0.00124	5	0
tropical	pointer	10	0.9825	[0.981, 0.983]	0.00124	5	0
tropical	pointer	100	0.9825	[0.981, 0.983]	0.00124	5	0
tropical	pointer	1000	0.9825	[0.981, 0.983]	0.00124	5	0
tropical	pointer	1e+04	0.9825	[0.981, 0.983]	0.00124	5	0
tropical	seq2seq	0	1	[1, 1]	0	5	0
tropical	seq2seq	0.01	1	[1, 1]	0	5	0
tropical	seq2seq	0.1	1	[1, 1]	0	5	0
tropical	seq2seq	1	1	[1, 1]	0	5	0
tropical	seq2seq	10	1	[1, 1]	0	5	0
tropical	seq2seq	100	1	[1, 1]	0	5	0
tropical	seq2seq	1000	1	[1, 1]	0	5	0
tropical	seq2seq	1e+04	1	[1, 1]	0	5	0
vanilla	pointer	0	0.9817	[0.98, 0.983]	0.00237	5	0
vanilla	pointer	0.01	0.981	[0.978, 0.983]	0.00376	5	0
vanilla	pointer	0.1	0.981	[0.978, 0.983]	0.00379	5	0
vanilla	pointer	1	0.9808	[0.977, 0.983]	0.0041	5	0
vanilla	pointer	10	0.9811	[0.978, 0.983]	0.00357	5	0
vanilla	pointer	100	0.9809	[0.978, 0.983]	0.00387	5	0
vanilla	pointer	1000	0.9807	[0.977, 0.983]	0.0045	5	0
vanilla	pointer	1e+04	0.9804	[0.976, 0.983]	0.00508	5	0
vanilla	seq2seq	0	1	[1, 1]	0	5	0
vanilla	seq2seq	0.01	1	[1, 1]	0	5	0
vanilla	seq2seq	0.1	1	[1, 1]	0	5	0
vanilla	seq2seq	1	1	[1, 1]	0	5	0
vanilla	seq2seq	10	1	[1, 1]	0	5	0
vanilla	seq2seq	100	1	[1, 1]	0	5	0
vanilla	seq2seq	1000	1	[1, 1]	0	5	0
vanilla	seq2seq	1e+04	1	[1, 1]	0	5	0

Table 10: `ood_val_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. rank: Wilcoxon signed-rank by seed, Holm.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	0	[-0.00259, 0]	1.0000
adaptive	pointer	0.1	0	[-0.00257, 0]	1.0000
adaptive	pointer	1	0	[-0.00265, 0]	1.0000
adaptive	pointer	10	0	[-0.00235, 0]	1.0000
adaptive	pointer	100	0	[-0.0026, 0]	1.0000
adaptive	pointer	1000	0	[-0.00373, 0]	1.0000

Table 10: `ood_val_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. rank: Wilcoxon signed-rank by seed, Holm.

`model_type`	`head`	`fishr_lambda`	difference	95% CI	p
adaptive	pointer	1e+04	0	[-0.00379, 0]	1.0000
adaptive	seq2seq	0.01	0	[0, 0]	–
adaptive	seq2seq	0.1	0	[0, 0]	–
adaptive	seq2seq	1	0	[0, 0]	–
adaptive	seq2seq	10	0	[0, 0]	–
adaptive	seq2seq	100	0	[0, 0]	–
adaptive	seq2seq	1000	0	[0, 0]	–
adaptive	seq2seq	1e+04	0	[0, 0]	–
tropical	pointer	0.01	0	[0, 0]	–
tropical	pointer	0.1	0	[0, 0]	–
tropical	pointer	1	0	[0, 0]	–
tropical	pointer	10	0	[0, 0]	–
tropical	pointer	100	0	[0, 0]	–
tropical	pointer	1000	0	[0, 0]	–
tropical	pointer	1e+04	0	[0, 0]	–
tropical	seq2seq	0.01	0	[0, 0]	–
tropical	seq2seq	0.1	0	[0, 0]	–
tropical	seq2seq	1	0	[0, 0]	–
tropical	seq2seq	10	0	[0, 0]	–
tropical	seq2seq	100	0	[0, 0]	–
tropical	seq2seq	1000	0	[0, 0]	–
tropical	seq2seq	1e+04	0	[0, 0]	–
vanilla	pointer	0.01	0	[-0.00205, 0]	1.0000
vanilla	pointer	0.1	0	[-0.00209, 0]	1.0000
vanilla	pointer	1	0	[-0.00252, 0]	1.0000
vanilla	pointer	10	0	[-0.00178, 0]	1.0000
vanilla	pointer	100	0	[-0.00221, 0]	1.0000
vanilla	pointer	1000	0	[-0.00309, 0]	1.0000
vanilla	pointer	1e+04	0	[-0.0039, 0]	1.0000
vanilla	seq2seq	0.01	0	[0, 0]	–
vanilla	seq2seq	0.1	0	[0, 0]	–
vanilla	seq2seq	1	0	[0, 0]	–
vanilla	seq2seq	10	0	[0, 0]	–
vanilla	seq2seq	100	0	[0, 0]	–
vanilla	seq2seq	1000	0	[0, 0]	–
vanilla	seq2seq	1e+04	0	[0, 0]	–

Table 11: `ood_val_f1` against log10 `fishr_lambda`: rank: Spearman rho, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	-0.03078	[-0.152, 0.0129]	0.8607	-0.0248	0.8598
adaptive	seq2seq	–	[–, –]	–	–	–
tropical	pointer	0	[0, 0]	1.0000	0	1.0000
tropical	seq2seq	–	[–, –]	–	–	–
vanilla	pointer	-0.02874	[-0.257, 0]	0.8698	-0.0248	0.8562
vanilla	seq2seq	–	[–, –]	–	–	–

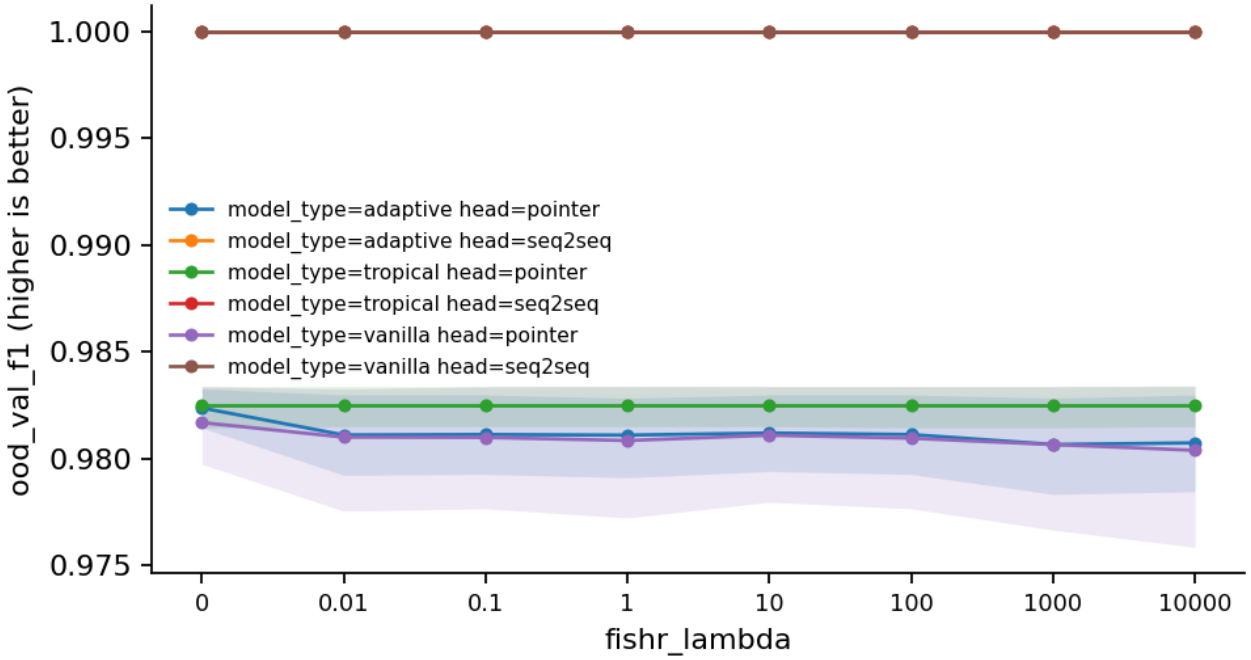


Figure 3: `ood_val_f1` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

Table 12: `ood_val_f1`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	8.035×10^{-4}	[-0.00143, 0.00303]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.001487	[-0.00264, 0.00562]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.001501	[-0.00267, 0.00567]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.001645	[-0.00292, 0.00621]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.001398	[-0.00248, 0.00528]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.001539	[-0.00273, 0.00581]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.001833	[-0.00326, 0.00692]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.002102	[-0.00373, 0.00794]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0	[–, –]	–
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0	[–, –]	–
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0	[–, –]	–
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0	[–, –]	–
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0	[–, –]	–

Table 12: `ood_val_f1`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0	[–, –]	–
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	0	[–, –]	–
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0	[–, –]	–
model_type: adaptive - vanilla head=pointer fishr_lambda=0	6.837×10^{-4}	[-0.0016, 0.00297]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	1.039×10^{-4}	[-0.00349, 0.0037]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	1.382×10^{-4}	[-0.00355, 0.00383]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1	2.540×10^{-4}	[-0.00381, 0.00432]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=10	1.056×10^{-4}	[-0.00319, 0.0034]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=100	1.757×10^{-4}	[-0.00366, 0.00401]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	3.736×10^{-6}	[-0.00354, 0.00355]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	3.478×10^{-4}	[-0.00337, 0.00406]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	0	[–, –]	–
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	-0.01764	[-0.0192, -0.0161]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	-0.0189	[-0.0219, -0.0159]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	-0.01888	[-0.0218, -0.016]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	-0.01891	[-0.0219, -0.0159]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	-0.01881	[-0.0216, -0.016]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	-0.01888	[-0.0218, -0.0159]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	-0.01935	[-0.023, -0.0157]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	-0.01927	[-0.023, -0.0156]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0	-0.01752	[-0.0191, -0.016]	1.0000

Table 12: `ood_val_f1`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	-0.01752	[-0.0191, -0.016]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	-0.01752	[-0.0191, -0.016]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1	-0.01752	[-0.0191, -0.016]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=10	-0.01752	[-0.0191, -0.016]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=100	-0.01752	[-0.0191, -0.016]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	-0.01752	[-0.0191, -0.016]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	-0.01752	[-0.0191, -0.016]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	-0.01832	[-0.0213, -0.0154]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	-0.019	[-0.0237, -0.0143]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	-0.01902	[-0.0237, -0.0143]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	-0.01916	[-0.0242, -0.0141]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	-0.01891	[-0.0234, -0.0145]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	-0.01906	[-0.0239, -0.0142]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	-0.01935	[-0.0249, -0.0138]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	-0.01962	[-0.0259, -0.0133]	1.0000

A.3 `ood_val_exact`

Table 13: `ood_val_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.9789	[0.977, 0.981]	0.00209	5	0
adaptive	pointer	0.01	0.9723	[0.964, 0.98]	0.0104	5	0
adaptive	pointer	0.1	0.9724	[0.964, 0.98]	0.0103	5	0
adaptive	pointer	1	0.9723	[0.964, 0.98]	0.0105	5	0
adaptive	pointer	10	0.9727	[0.965, 0.98]	0.00983	5	0
adaptive	pointer	100	0.9724	[0.965, 0.98]	0.0103	5	0
adaptive	pointer	1000	0.97	[0.959, 0.98]	0.0138	5	0
adaptive	pointer	1e+04	0.9703	[0.959, 0.98]	0.0139	5	0
adaptive	seq2seq	0	1	[1, 1]	0	5	0
adaptive	seq2seq	0.01	1	[1, 1]	0	5	0
adaptive	seq2seq	0.1	1	[1, 1]	0	5	0
adaptive	seq2seq	1	1	[1, 1]	0	5	0
adaptive	seq2seq	10	1	[1, 1]	0	5	0
adaptive	seq2seq	100	1	[1, 1]	0	5	0
adaptive	seq2seq	1000	1	[1, 1]	0	5	0
adaptive	seq2seq	1e+04	1	[1, 1]	0	5	0
tropical	pointer	0	0.9796	[0.978, 0.981]	0.00181	5	0
tropical	pointer	0.01	0.9796	[0.978, 0.981]	0.00181	5	0

Table 13: `ood_val_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
tropical	pointer	0.1	0.9796	[0.978, 0.981]	0.00181	5	0
tropical	pointer	1	0.9796	[0.978, 0.981]	0.00181	5	0
tropical	pointer	10	0.9796	[0.978, 0.981]	0.00181	5	0
tropical	pointer	100	0.9796	[0.978, 0.981]	0.00181	5	0
tropical	pointer	1000	0.9796	[0.978, 0.981]	0.00181	5	0
tropical	pointer	1e+04	0.9796	[0.978, 0.981]	0.00181	5	0
tropical	seq2seq	0	1	[1, 1]	0	5	0
tropical	seq2seq	0.01	1	[1, 1]	0	5	0
tropical	seq2seq	0.1	1	[1, 1]	0	5	0
tropical	seq2seq	1	1	[1, 1]	0	5	0
tropical	seq2seq	10	1	[1, 1]	0	5	0
tropical	seq2seq	100	1	[1, 1]	0	5	0
tropical	seq2seq	1000	1	[1, 1]	0	5	0
tropical	seq2seq	1e+04	1	[1, 1]	0	5	0
vanilla	pointer	0	0.9753	[0.966, 0.981]	0.0104	5	0
vanilla	pointer	0.01	0.9716	[0.955, 0.981]	0.0184	5	0
vanilla	pointer	0.1	0.9716	[0.955, 0.981]	0.0186	5	0
vanilla	pointer	1	0.9708	[0.953, 0.981]	0.0202	5	0
vanilla	pointer	10	0.9721	[0.957, 0.981]	0.0173	5	0
vanilla	pointer	100	0.9713	[0.954, 0.981]	0.0191	5	0
vanilla	pointer	1000	0.9698	[0.95, 0.981]	0.0224	5	0
vanilla	pointer	1e+04	0.9684	[0.945, 0.981]	0.0256	5	0
vanilla	seq2seq	0	1	[1, 1]	0	5	0
vanilla	seq2seq	0.01	1	[1, 1]	0	5	0
vanilla	seq2seq	0.1	1	[1, 1]	0	5	0
vanilla	seq2seq	1	1	[1, 1]	0	5	0
vanilla	seq2seq	10	1	[1, 1]	0	5	0
vanilla	seq2seq	100	1	[1, 1]	0	5	0
vanilla	seq2seq	1000	1	[1, 1]	0	5	0
vanilla	seq2seq	1e+04	1	[1, 1]	0	5	0

Table 14: `ood_val_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. rank: Wilcoxon signed-rank by seed, Holm.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	0	[-0.0134, 0]	1.0000
adaptive	pointer	0.1	0	[-0.0133, 0]	1.0000
adaptive	pointer	1	0	[-0.0137, 0]	1.0000
adaptive	pointer	10	0	[-0.0126, 0]	1.0000
adaptive	pointer	100	0	[-0.0134, 0]	1.0000
adaptive	pointer	1000	0	[-0.0196, 0]	1.0000
adaptive	pointer	1e+04	0	[-0.0201, 0]	1.0000
adaptive	seq2seq	0.01	0	[0, 0]	–
adaptive	seq2seq	0.1	0	[0, 0]	–
adaptive	seq2seq	1	0	[0, 0]	–
adaptive	seq2seq	10	0	[0, 0]	–
adaptive	seq2seq	100	0	[0, 0]	–
adaptive	seq2seq	1000	0	[0, 0]	–
adaptive	seq2seq	1e+04	0	[0, 0]	–
tropical	pointer	0.01	0	[0, 0]	–
tropical	pointer	0.1	0	[0, 0]	–
tropical	pointer	1	0	[0, 0]	–
tropical	pointer	10	0	[0, 0]	–

Table 14: `ood_val_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. rank: Wilcoxon signed-rank by seed, Holm.

`model_type`	`head`	`fishr_lambda`	difference	95% CI	p
tropical	pointer	100	0	[0, 0]	–
tropical	pointer	1000	0	[0, 0]	–
tropical	pointer	1e+04	0	[0, 0]	–
tropical	seq2seq	0.01	0	[0, 0]	–
tropical	seq2seq	0.1	0	[0, 0]	–
tropical	seq2seq	1	0	[0, 0]	–
tropical	seq2seq	10	0	[0, 0]	–
tropical	seq2seq	100	0	[0, 0]	–
tropical	seq2seq	1000	0	[0, 0]	–
tropical	seq2seq	1e+04	0	[0, 0]	–
vanilla	pointer	0.01	0	[-0.0109, 0]	1.0000
vanilla	pointer	0.1	0	[-0.0111, 0]	1.0000
vanilla	pointer	1	0	[-0.0133, 0]	1.0000
vanilla	pointer	10	0	[-0.00936, 0]	1.0000
vanilla	pointer	100	0	[-0.0118, 0]	1.0000
vanilla	pointer	1000	0	[-0.0163, 0]	1.0000
vanilla	pointer	1e+04	0	[-0.0205, 0]	1.0000
vanilla	seq2seq	0.01	0	[0, 0]	–
vanilla	seq2seq	0.1	0	[0, 0]	–
vanilla	seq2seq	1	0	[0, 0]	–
vanilla	seq2seq	10	0	[0, 0]	–
vanilla	seq2seq	100	0	[0, 0]	–
vanilla	seq2seq	1000	0	[0, 0]	–
vanilla	seq2seq	1e+04	0	[0, 0]	–

Table 15: `ood_val_exact` against log10 `fishr_lambda`: rank: Spearman rho, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	-0.03292	[-0.158, 0.0129]	0.8511	-0.0305	0.8254
adaptive	seq2seq	–	[–, –]	–	–	–
tropical	pointer	0	[0, 0]	1.0000	0	1.0000
tropical	seq2seq	–	[–, –]	–	–	–
vanilla	pointer	-0.02874	[-0.257, 0]	0.8698	-0.0248	0.8562
vanilla	seq2seq	–	[–, –]	–	–	–

Table 16: `ood_val_exact`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.00432	[-0.00767, 0.0163]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.00796	[-0.0141, 0.0301]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.00802	[-0.0142, 0.0303]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.00876	[-0.0156, 0.0331]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.00744	[-0.0132, 0.0281]	1.0000

Table 16: `ood_val_exact`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.00824	[-0.0146, 0.0311]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.00976	[-0.0173, 0.0369]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.01116	[-0.0198, 0.0421]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0	[–, –]	–
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0	[–, –]	–
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0	[–, –]	–
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0	[–, –]	–
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0	[–, –]	–
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0	[–, –]	–
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	0	[–, –]	–
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0	[–, –]	–
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.00368	[-0.00857, 0.0159]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	6.800×10^{-4}	[-0.0181, 0.0194]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	8.400×10^{-4}	[-0.0184, 0.02]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1	0.00144	[-0.0197, 0.0226]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=10	5.600×10^{-4}	[-0.0165, 0.0176]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=100	0.00108	[-0.0189, 0.0211]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	2.000×10^{-4}	[-0.0183, 0.0187]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	0.0019	[-0.0176, 0.0214]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	0	[–, –]	–
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	-0.02106	[-0.0237, -0.0185]	1.0000

Table 16: `ood_val_exact`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	-0.0277	[-0.0406, -0.0148]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	-0.0276	[-0.0403, -0.0149]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	-0.02774	[-0.0407, -0.0147]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	-0.0273	[-0.0395, -0.0151]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	-0.02758	[-0.0403, -0.0148]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	-0.02998	[-0.0471, -0.0129]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	-0.02968	[-0.0469, -0.0124]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0	-0.02042	[-0.0227, -0.0182]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	-0.02042	[-0.0227, -0.0182]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	-0.02042	[-0.0227, -0.0182]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1	-0.02042	[-0.0227, -0.0182]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=10	-0.02042	[-0.0227, -0.0182]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=100	-0.02042	[-0.0227, -0.0182]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	-0.02042	[-0.0227, -0.0182]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	-0.02474	[-0.0376, -0.0119]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	-0.02838	[-0.0513, -0.0055]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	-0.02844	[-0.0515, -0.00539]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	-0.02918	[-0.0543, -0.00409]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	-0.02786	[-0.0493, -0.00642]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	-0.02866	[-0.0523, -0.005]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	-0.03018	[-0.058, -0.00232]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	-0.03158	[-0.0633, 1.58×10^{-4}]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04			

A.4 `ood_noise_f1`

Table 17: `ood_noise_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.8918	[0.889, 0.897]	0.00546	5	0
adaptive	pointer	0.01	0.8869	[0.883, 0.89]	0.00449	5	0

Table 17: `ood_noise_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	pointer	0.1	0.8869	[0.883, 0.89]	0.00445	5	0
adaptive	pointer	1	0.8869	[0.883, 0.89]	0.0045	5	0
adaptive	pointer	10	0.887	[0.883, 0.89]	0.00439	5	0
adaptive	pointer	100	0.887	[0.883, 0.89]	0.00431	5	0
adaptive	pointer	1000	0.8871	[0.884, 0.89]	0.00402	5	0
adaptive	pointer	1e+04	0.8868	[0.883, 0.89]	0.00492	5	0
adaptive	seq2seq	0	1	[1, 1]	0	5	0
adaptive	seq2seq	0.01	1	[1, 1]	0	5	0
adaptive	seq2seq	0.1	1	[1, 1]	0	5	0
adaptive	seq2seq	1	1	[1, 1]	0	5	0
adaptive	seq2seq	10	1	[1, 1]	0	5	0
adaptive	seq2seq	100	1	[1, 1]	0	5	0
adaptive	seq2seq	1000	1	[1, 1]	0	5	0
adaptive	seq2seq	1e+04	1	[1, 1]	0	5	0
tropical	pointer	0	0.8893	[0.888, 0.89]	0.00102	5	0
tropical	pointer	0.01	0.8894	[0.889, 0.89]	9.29×10^{-4}	5	0
tropical	pointer	0.1	0.889	[0.888, 0.89]	0.00166	5	0
tropical	pointer	1	0.8937	[0.889, 0.902]	0.0089	5	0
tropical	pointer	10	0.8894	[0.889, 0.89]	9.71×10^{-4}	5	0
tropical	pointer	100	0.8893	[0.888, 0.89]	0.00109	5	0
tropical	pointer	1000	0.8894	[0.889, 0.89]	0.00101	5	0
tropical	pointer	1e+04	0.8895	[0.889, 0.89]	8.27×10^{-4}	5	0
tropical	seq2seq	0	0.9928	[0.986, 0.999]	0.0081	5	0
tropical	seq2seq	0.01	0.9987	[0.996, 1]	0.003	5	0
tropical	seq2seq	0.1	0.9987	[0.996, 1]	0.003	5	0
tropical	seq2seq	1	0.9987	[0.996, 1]	0.003	5	0
tropical	seq2seq	10	0.9987	[0.996, 1]	0.003	5	0
tropical	seq2seq	100	0.9987	[0.996, 1]	0.003	5	0
tropical	seq2seq	1000	0.9987	[0.996, 1]	0.003	5	0
tropical	seq2seq	1e+04	0.9987	[0.996, 1]	0.003	5	0
vanilla	pointer	0	0.8877	[0.884, 0.89]	0.00403	5	0
vanilla	pointer	0.01	0.887	[0.882, 0.89]	0.00546	5	0
vanilla	pointer	0.1	0.887	[0.882, 0.89]	0.00549	5	0
vanilla	pointer	1	0.887	[0.882, 0.89]	0.00539	5	0
vanilla	pointer	10	0.8873	[0.883, 0.89]	0.00486	5	0
vanilla	pointer	100	0.8871	[0.882, 0.89]	0.00521	5	0
vanilla	pointer	1000	0.8868	[0.881, 0.89]	0.00602	5	0
vanilla	pointer	1e+04	0.8867	[0.881, 0.89]	0.00624	5	0
vanilla	seq2seq	0	1	[1, 1]	0	5	0
vanilla	seq2seq	0.01	1	[1, 1]	0	5	0
vanilla	seq2seq	0.1	1	[1, 1]	0	5	0
vanilla	seq2seq	1	1	[1, 1]	0	5	0
vanilla	seq2seq	10	1	[1, 1]	0	5	0
vanilla	seq2seq	100	1	[1, 1]	0	5	0
vanilla	seq2seq	1000	1	[1, 1]	0	5	0
vanilla	seq2seq	1e+04	1	[1, 1]	0	5	0

Table 18: `ood_noise_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. rank: Wilcoxon signed-rank by seed, Holm.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	0	[-0.0107, 0]	1.0000
adaptive	pointer	0.1	0	[-0.0107, 0]	1.0000

Table 18: `ood_noise_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. rank: Wilcoxon signed-rank by seed, Holm.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	1	0	[-0.0107, 0]	1.0000
adaptive	pointer	10	0	[-0.0106, 0]	1.0000
adaptive	pointer	100	0	[-0.0107, 0]	1.0000
adaptive	pointer	1000	0	[-0.0106, 0]	1.0000
adaptive	pointer	1e+04	0	[-0.0107, 0]	1.0000
adaptive	seq2seq	0.01	0	[0, 0]	–
adaptive	seq2seq	0.1	0	[0, 0]	–
adaptive	seq2seq	1	0	[0, 0]	–
adaptive	seq2seq	10	0	[0, 0]	–
adaptive	seq2seq	100	0	[0, 0]	–
adaptive	seq2seq	1000	0	[0, 0]	–
adaptive	seq2seq	1e+04	0	[0, 0]	–
tropical	pointer	0.01	0	$[-1.83 \times 10^{-4}, 5.76 \times 10^{-4}]$	1.0000
tropical	pointer	0.1	0	$[-0.00132, 4.36 \times 10^{-4}]$	1.0000
tropical	pointer	1	6.813×10^{-5}	$[4.55 \times 10^{-6}, 0.0129]$	1.0000
tropical	pointer	10	0	$[-2.59 \times 10^{-4}, 5.78 \times 10^{-4}]$	1.0000
tropical	pointer	100	0	$[-4.45 \times 10^{-4}, 5.27 \times 10^{-4}]$	1.0000
tropical	pointer	1000	0	$[-3.32 \times 10^{-4}, 6.21 \times 10^{-4}]$	1.0000
tropical	pointer	1e+04	6.784×10^{-5}	$[-4.53 \times 10^{-5}, 6.14 \times 10^{-4}]$	1.0000
tropical	seq2seq	0.01	0.005907	[0.00118, 0.0106]	1.0000
tropical	seq2seq	0.1	0.005907	[0.00118, 0.0106]	1.0000
tropical	seq2seq	1	0.005907	[0.00118, 0.0106]	1.0000
tropical	seq2seq	10	0.005907	[0.00118, 0.0106]	1.0000
tropical	seq2seq	100	0.005907	[0.00118, 0.0106]	1.0000
tropical	seq2seq	1000	0.005907	[0.00118, 0.0106]	1.0000
tropical	seq2seq	1e+04	0.005907	[0.00118, 0.0106]	1.0000
vanilla	pointer	0.01	0	[-0.00193, 0]	1.0000
vanilla	pointer	0.1	0	[-0.00197, 0]	1.0000
vanilla	pointer	1	0	[-0.00185, 0]	1.0000
vanilla	pointer	10	0	[-0.00113, 0]	1.0000
vanilla	pointer	100	0	[-0.0016, 0]	1.0000
vanilla	pointer	1000	0	[-0.0027, 0]	1.0000
vanilla	pointer	1e+04	0	[-0.00299, 0]	1.0000
vanilla	seq2seq	0.01	0	[0, 0]	–
vanilla	seq2seq	0.1	0	[0, 0]	–
vanilla	seq2seq	1	0	[0, 0]	–
vanilla	seq2seq	10	0	[0, 0]	–
vanilla	seq2seq	100	0	[0, 0]	–
vanilla	seq2seq	1000	0	[0, 0]	–
vanilla	seq2seq	1e+04	0	[0, 0]	–

Table 19: `ood_noise_f1` against log10 `fishr_lambda`: rank: Spearman rho, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

model_type	head	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	0.004294	[-0.0129, 0.0514]	0.9805	0.00381	0.9844
adaptive	seq2seq	–	[–, –]	–	–	–
tropical	pointer	-0.01279	[-0.155, 0.161]	0.9419	-0.0114	0.9394
tropical	seq2seq	0	[0, 0]	1.0000	0	1.0000
vanilla	pointer	-0.01437	[-0.129, 0]	0.9347	-0.00952	0.9508
vanilla	seq2seq	–	[–, –]	–	–	–

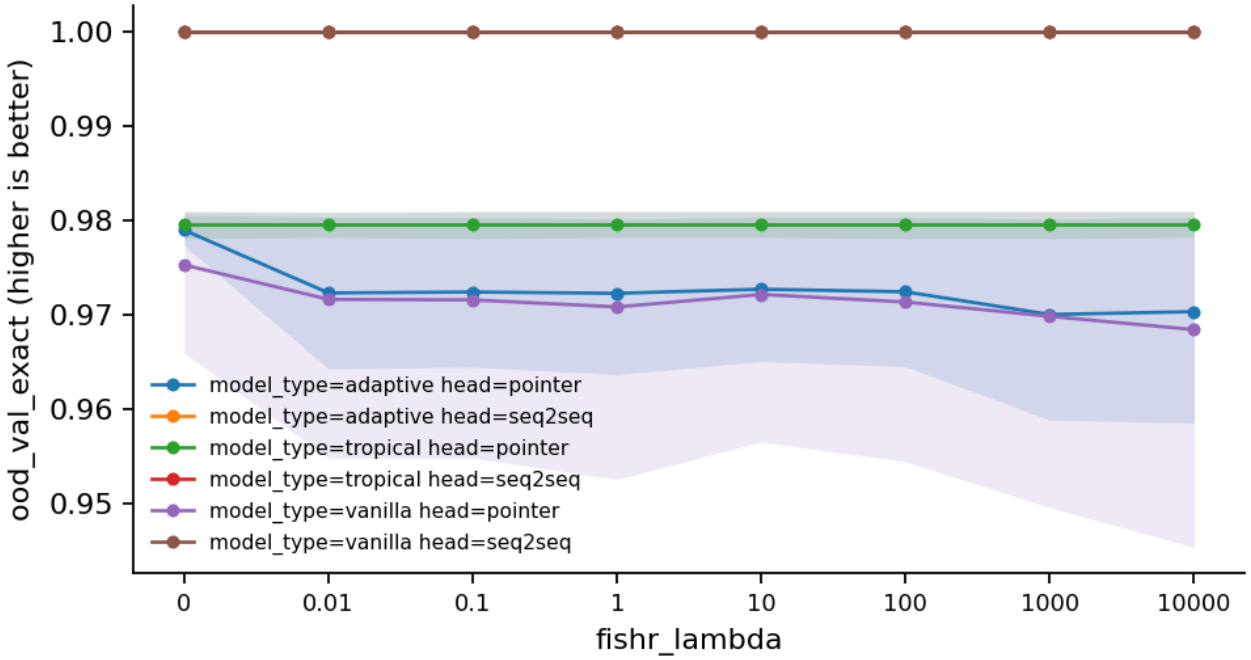


Figure 4: `ood_val_exact` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

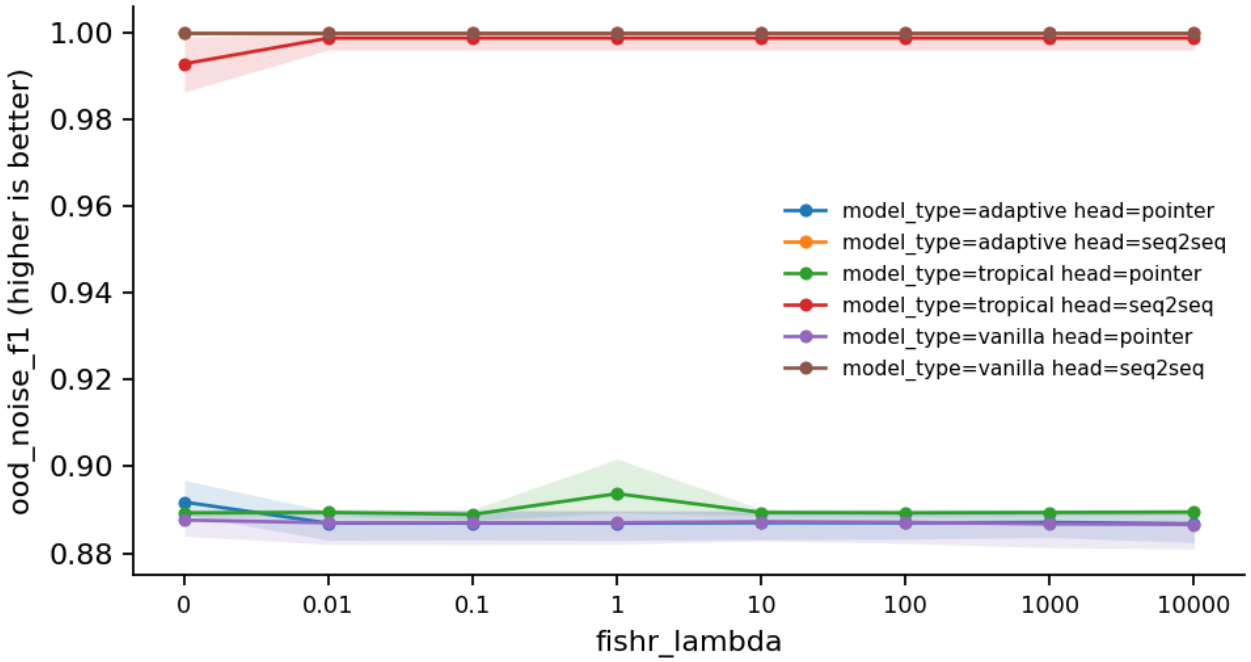


Figure 5: `ood_noise_f1` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

Table 20: `ood_noise_f1`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.001636	[-0.00391, 0.00718]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.002408	[-0.00482, 0.00963]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.001955	[-0.00573, 0.00964]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.006687	[-0.00527, 0.0186]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.002117	[-0.00438, 0.00862]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.002187	[-0.00482, 0.00919]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.002635	[-0.00532, 0.0106]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.002839	[-0.00531, 0.011]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	-0.00725	[-0.0173, 0.00281]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	-0.00134	[-0.00506, 0.00238]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	-0.00134	[-0.00506, 0.00238]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	-0.00134	[-0.00506, 0.00238]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	-0.00134	[-0.00506, 0.00238]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	-0.00134	[-0.00506, 0.00238]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	-0.00134	[-0.00506, 0.00238]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	-0.00134	[-0.00506, 0.00238]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.004142	[-0.0031, 0.0114]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	-7.061×10^{-5}	[-0.00192, 0.00178]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	-5.307×10^{-5}	[-0.002, 0.0019]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1	-1.174×10^{-4}	[-0.00194, 0.00171]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=10	-2.617×10^{-4}	[-0.00163, 0.00111]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=100	-1.239×10^{-4}	[-0.002, 0.00175]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	3.598×10^{-4}	[-0.00272, 0.00344]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	1.405×10^{-4}	[-0.00193, 0.00221]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	0	[–, –]	–

Table 20: `ood_noise_f1`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	0	[–, –]	–
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	-0.1082	[-0.115, -0.101]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	-0.1131	[-0.119, -0.107]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	-0.1131	[-0.119, -0.108]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	-0.1131	[-0.119, -0.107]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	-0.113	[-0.118, -0.108]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	-0.113	[-0.118, -0.108]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	-0.1129	[-0.118, -0.108]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	-0.1132	[-0.119, -0.107]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0	-0.1035	[-0.113, -0.0936]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	-0.1092	[-0.112, -0.106]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	-0.1097	[-0.112, -0.108]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1	-0.1049	[-0.12, -0.0902]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=10	-0.1093	[-0.112, -0.106]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=100	-0.1093	[-0.112, -0.107]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	-0.1093	[-0.112, -0.106]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	-0.1092	[-0.112, -0.106]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	-0.1123	[-0.117, -0.107]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	-0.113	[-0.12, -0.106]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	-0.113	[-0.12, -0.106]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	-0.113	[-0.12, -0.106]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	-0.1127	[-0.119, -0.107]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	-0.1129	[-0.119, -0.106]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	-0.1132	[-0.121, -0.106]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	-0.1133	[-0.121, -0.106]	1.0000

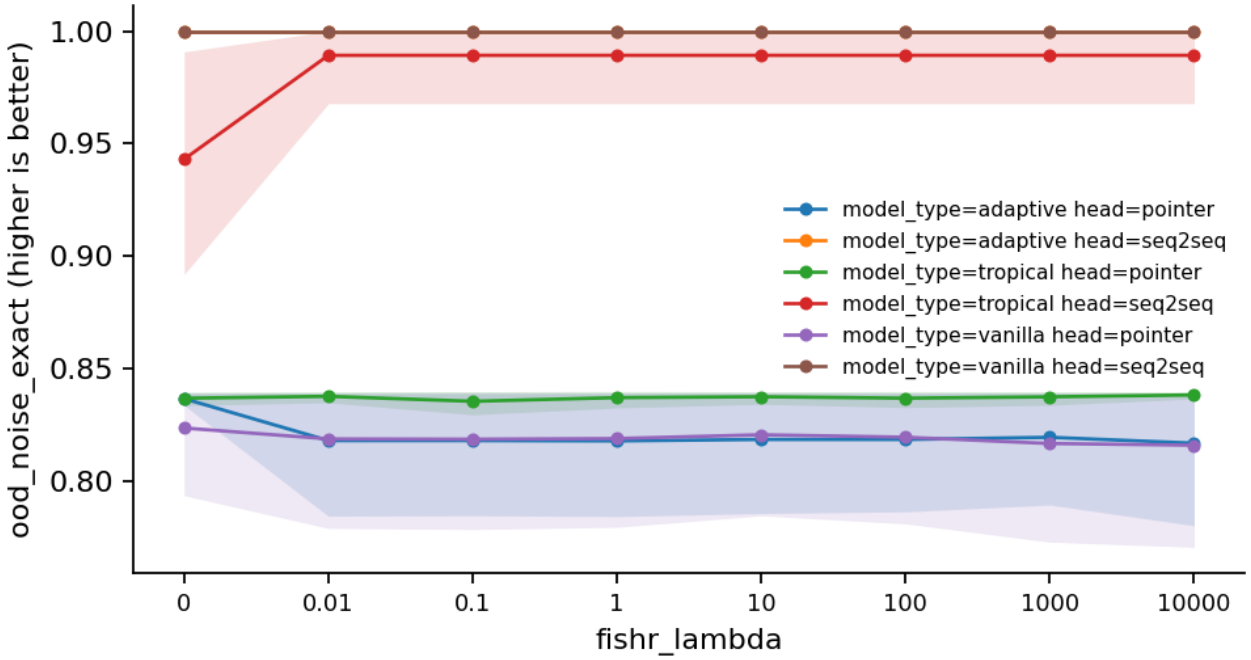


Figure 6: `ood_noise_exact` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

A.5 `ood_noise_exact`

Table 21: `ood_noise_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.8365	[0.834, 0.839]	0.00344	5	0
adaptive	pointer	0.01	0.8178	[0.784, 0.839]	0.0367	5	0
adaptive	pointer	0.1	0.8178	[0.784, 0.839]	0.0365	5	0
adaptive	pointer	1	0.8177	[0.784, 0.839]	0.0367	5	0
adaptive	pointer	10	0.8183	[0.785, 0.839]	0.036	5	0
adaptive	pointer	100	0.8183	[0.786, 0.839]	0.0351	5	0
adaptive	pointer	1000	0.8193	[0.789, 0.839]	0.0329	5	0
adaptive	pointer	1e+04	0.8167	[0.78, 0.839]	0.0401	5	0
adaptive	seq2seq	0	1	[1, 1]	0	5	0
adaptive	seq2seq	0.01	1	[1, 1]	0	5	0
adaptive	seq2seq	0.1	1	[1, 1]	0	5	0
adaptive	seq2seq	1	1	[1, 1]	0	5	0
adaptive	seq2seq	10	1	[1, 1]	0	5	0
adaptive	seq2seq	100	1	[1, 1]	0	5	0
adaptive	seq2seq	1000	1	[1, 1]	0	5	0
adaptive	seq2seq	1e+04	1	[1, 1]	0	5	0
tropical	pointer	0	0.8367	[0.833, 0.839]	0.00375	5	0
tropical	pointer	0.01	0.8375	[0.835, 0.84]	0.00328	5	0
tropical	pointer	0.1	0.8353	[0.829, 0.84]	0.00654	5	0
tropical	pointer	1	0.8369	[0.832, 0.84]	0.00503	5	0
tropical	pointer	10	0.8373	[0.834, 0.84]	0.00369	5	0
tropical	pointer	100	0.8367	[0.833, 0.84]	0.0047	5	0
tropical	pointer	1000	0.8373	[0.834, 0.84]	0.00418	5	0
tropical	pointer	1e+04	0.8381	[0.836, 0.84]	0.00211	5	0
tropical	seq2seq	0	0.943	[0.892, 0.991]	0.064	5	0

Table 21: `ood_noise_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
tropical	seq2seq	0.01	0.9893	[0.968, 1]	0.024	5	0
tropical	seq2seq	0.1	0.9893	[0.968, 1]	0.024	5	0
tropical	seq2seq	1	0.9893	[0.968, 1]	0.024	5	0
tropical	seq2seq	10	0.9893	[0.968, 1]	0.024	5	0
tropical	seq2seq	100	0.9893	[0.968, 1]	0.024	5	0
tropical	seq2seq	1000	0.9893	[0.968, 1]	0.024	5	0
tropical	seq2seq	1e+04	0.9893	[0.968, 1]	0.024	5	0
vanilla	pointer	0	0.8234	[0.793, 0.839]	0.0336	5	0
vanilla	pointer	0.01	0.8186	[0.779, 0.839]	0.0444	5	0
vanilla	pointer	0.1	0.8184	[0.778, 0.839]	0.0447	5	0
vanilla	pointer	1	0.8187	[0.779, 0.839]	0.0441	5	0
vanilla	pointer	10	0.8204	[0.784, 0.839]	0.0403	5	0
vanilla	pointer	100	0.8193	[0.781, 0.839]	0.0427	5	0
vanilla	pointer	1000	0.8165	[0.773, 0.839]	0.049	5	0
vanilla	pointer	1e+04	0.8158	[0.77, 0.839]	0.0507	5	0
vanilla	seq2seq	0	1	[1, 1]	0	5	0
vanilla	seq2seq	0.01	1	[1, 1]	0	5	0
vanilla	seq2seq	0.1	1	[1, 1]	0	5	0
vanilla	seq2seq	1	1	[1, 1]	0	5	0
vanilla	seq2seq	10	1	[1, 1]	0	5	0
vanilla	seq2seq	100	1	[1, 1]	0	5	0
vanilla	seq2seq	1000	1	[1, 1]	0	5	0
vanilla	seq2seq	1e+04	1	[1, 1]	0	5	0

Table 22: `ood_noise_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. rank: Wilcoxon signed-rank by seed, Holm.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	0	[-0.0464, 0]	1.0000
adaptive	pointer	0.1	0	[-0.0461, 0]	1.0000
adaptive	pointer	1	0	[-0.0464, 0]	1.0000
adaptive	pointer	10	0	[-0.0454, 0]	1.0000
adaptive	pointer	100	0	[-0.0442, 0]	1.0000
adaptive	pointer	1000	0	[-0.0411, 0]	1.0000
adaptive	pointer	1e+04	0	[-0.051, 0]	1.0000
adaptive	seq2seq	0.01	0	[0, 0]	–
adaptive	seq2seq	0.1	0	[0, 0]	–
adaptive	seq2seq	1	0	[0, 0]	–
adaptive	seq2seq	10	0	[0, 0]	–
adaptive	seq2seq	100	0	[0, 0]	–
adaptive	seq2seq	1000	0	[0, 0]	–
adaptive	seq2seq	1e+04	0	[0, 0]	–
tropical	pointer	0.01	0	[-0.00144, 0.00404]	1.0000
tropical	pointer	0.1	0	[-0.00588, 0.00315]	1.0000
tropical	pointer	1	-1.000×10^{-4}	[-0.0038, 0.0044]	1.0000
tropical	pointer	10	0	[-0.00204, 0.00406]	1.0000
tropical	pointer	100	0	[-0.0034, 0.00369]	1.0000
tropical	pointer	1000	0	[-0.0027, 0.00444]	1.0000
tropical	pointer	1e+04	6.000×10^{-4}	$[-4.00 \times 10^{-4}$, 0.00438]	1.0000
tropical	seq2seq	0.01	0.0458	[0.00916, 0.0833]	1.0000
tropical	seq2seq	0.1	0.0458	[0.00916, 0.0833]	1.0000
tropical	seq2seq	1	0.0458	[0.00916, 0.0833]	1.0000
tropical	seq2seq	10	0.0458	[0.00916, 0.0833]	1.0000

Table 22: `ood_noise_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. rank: Wilcoxon signed-rank by seed, Holm.

model_type	head	fishr_lambda	difference	95% CI	p
tropical	seq2seq	100	0.0458	[0.00916, 0.0833]	1.0000
tropical	seq2seq	1000	0.0458	[0.00916, 0.0833]	1.0000
tropical	seq2seq	1e+04	0.0458	[0.00916, 0.0833]	1.0000
vanilla	pointer	0.01	0	[-0.0146, 0]	1.0000
vanilla	pointer	0.1	0	[-0.0149, 0]	1.0000
vanilla	pointer	1	0	[-0.0141, 0]	1.0000
vanilla	pointer	10	0	[-0.009, 0]	1.0000
vanilla	pointer	100	0	[-0.0123, 0]	1.0000
vanilla	pointer	1000	0	[-0.0207, 0]	1.0000
vanilla	pointer	1e+04	0	[-0.023, 0]	1.0000
vanilla	seq2seq	0.01	0	[0, 0]	–
vanilla	seq2seq	0.1	0	[0, 0]	–
vanilla	seq2seq	1	0	[0, 0]	–
vanilla	seq2seq	10	0	[0, 0]	–
vanilla	seq2seq	100	0	[0, 0]	–
vanilla	seq2seq	1000	0	[0, 0]	–
vanilla	seq2seq	1e+04	0	[0, 0]	–

Table 23: `ood_noise_exact` against log10 `fishr_lambda`: rank: Spearman rho, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

model_type	head	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	0.004294	[-0.0129, 0.0514]	0.9805	0.00381	0.9844
adaptive	seq2seq	–	[–, –]	–	–	–
tropical	pointer	0.03553	[-0.139, 0.195]	0.8394	0.0267	0.8438
tropical	seq2seq	0	[0, 0]	1.0000	0	1.0000
vanilla	pointer	-0.01437	[-0.129, 0]	0.9347	-0.00952	0.9454
vanilla	seq2seq	–	[–, –]	–	–	–

Table 24: `ood_noise_exact`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.01326	[-0.0311, 0.0577]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.01896	[-0.0382, 0.0762]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.01686	[-0.0425, 0.0762]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.01822	[-0.039, 0.0754]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.01692	[-0.0353, 0.0691]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.01738	[-0.0384, 0.0731]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.02078	[-0.0423, 0.0838]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.02238	[-0.0421, 0.0869]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	-0.05696	[-0.136, 0.0225]	1.0000

Table 24: `ood_noise_exact`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	-0.01072	[-0.0405, 0.019]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	-0.01072	[-0.0405, 0.019]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	-0.01072	[-0.0405, 0.019]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	-0.01072	[-0.0405, 0.019]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	-0.01072	[-0.0405, 0.019]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	-0.01072	[-0.0405, 0.019]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	-0.01072	[-0.0405, 0.019]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.01306	[-0.025, 0.0511]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	-7.400×10^{-4}	[-0.0154, 0.0139]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	-6.200×10^{-4}	[-0.016, 0.0147]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1	-0.00104	[-0.0156, 0.0136]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=10	-0.0021	[-0.0136, 0.0094]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=100	-0.001	[-0.0161, 0.0141]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.00276	[-0.0213, 0.0268]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	9.000×10^{-4}	[-0.0157, 0.0175]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	0	[–, –]	–
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	-0.1635	[-0.168, -0.159]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	-0.1822	[-0.228, -0.137]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	-0.1822	[-0.227, -0.137]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	-0.1823	[-0.228, -0.137]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	-0.1817	[-0.226, -0.137]	1.0000

Table 24: `ood_noise_exact`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	-0.1817	[-0.225, -0.138]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	-0.1807	[-0.222, -0.14]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	-0.1833	[-0.233, -0.134]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0	-0.1064	[-0.184, -0.0284]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	-0.1518	[-0.178, -0.126]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	-0.154	[-0.177, -0.131]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1	-0.1523	[-0.176, -0.129]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=10	-0.1519	[-0.177, -0.126]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=100	-0.1526	[-0.177, -0.128]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	-0.152	[-0.177, -0.127]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	-0.1511	[-0.179, -0.123]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	-0.1766	[-0.218, -0.135]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	-0.1814	[-0.237, -0.126]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	-0.1816	[-0.237, -0.126]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	-0.1813	[-0.236, -0.127]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	-0.1796	[-0.23, -0.13]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	-0.1807	[-0.234, -0.128]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	-0.1835	[-0.244, -0.123]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	-0.1842	[-0.247, -0.121]	1.0000

A.6 `id_f1`

Table 25: `id_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.9861	[0.985, 0.987]	7.86×10^{-4}	5	0
adaptive	pointer	0.01	0.9861	[0.985, 0.987]	7.86×10^{-4}	5	0
adaptive	pointer	0.1	0.9861	[0.985, 0.987]	7.86×10^{-4}	5	0
adaptive	pointer	1	0.9861	[0.985, 0.987]	7.86×10^{-4}	5	0
adaptive	pointer	10	0.9861	[0.985, 0.987]	7.86×10^{-4}	5	0
adaptive	pointer	100	0.9861	[0.985, 0.987]	7.86×10^{-4}	5	0
adaptive	pointer	1000	0.9861	[0.985, 0.987]	7.86×10^{-4}	5	0
adaptive	pointer	1e+04	0.9861	[0.985, 0.987]	7.86×10^{-4}	5	0
adaptive	seq2seq	0	1	[1, 1]	0	5	0
adaptive	seq2seq	0.01	1	[1, 1]	0	5	0

Table 25: `id_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	seq2seq	0.1	1	[1, 1]	0	5	0
adaptive	seq2seq	1	1	[1, 1]	0	5	0
adaptive	seq2seq	10	1	[1, 1]	0	5	0
adaptive	seq2seq	100	1	[1, 1]	0	5	0
adaptive	seq2seq	1000	1	[1, 1]	0	5	0
adaptive	seq2seq	1e+04	1	[1, 1]	0	5	0
tropical	pointer	0	0.9861	[0.985, 0.987]	7.86×10^{-4}	5	0
tropical	pointer	0.01	0.9861	[0.985, 0.987]	7.86×10^{-4}	5	0
tropical	pointer	0.1	0.9861	[0.985, 0.987]	7.86×10^{-4}	5	0
tropical	pointer	1	0.9861	[0.985, 0.987]	7.86×10^{-4}	5	0
tropical	pointer	10	0.9861	[0.985, 0.987]	7.86×10^{-4}	5	0
tropical	pointer	100	0.9861	[0.985, 0.987]	7.86×10^{-4}	5	0
tropical	pointer	1000	0.9861	[0.985, 0.987]	7.86×10^{-4}	5	0
tropical	pointer	1e+04	0.9861	[0.985, 0.987]	7.86×10^{-4}	5	0
tropical	seq2seq	0	1	[1, 1]	0	5	0
tropical	seq2seq	0.01	1	[1, 1]	0	5	0
tropical	seq2seq	0.1	1	[1, 1]	0	5	0
tropical	seq2seq	1	1	[1, 1]	0	5	0
tropical	seq2seq	10	1	[1, 1]	0	5	0
tropical	seq2seq	100	1	[1, 1]	0	5	0
tropical	seq2seq	1000	1	[1, 1]	0	5	0
tropical	seq2seq	1e+04	1	[1, 1]	0	5	0
vanilla	pointer	0	0.9861	[0.985, 0.987]	7.86×10^{-4}	5	0
vanilla	pointer	0.01	0.9861	[0.985, 0.987]	7.86×10^{-4}	5	0
vanilla	pointer	0.1	0.9861	[0.985, 0.987]	7.86×10^{-4}	5	0
vanilla	pointer	1	0.9861	[0.985, 0.987]	7.86×10^{-4}	5	0
vanilla	pointer	10	0.9861	[0.985, 0.987]	7.86×10^{-4}	5	0
vanilla	pointer	100	0.9861	[0.985, 0.987]	7.86×10^{-4}	5	0
vanilla	pointer	1000	0.9861	[0.985, 0.987]	7.86×10^{-4}	5	0
vanilla	pointer	1e+04	0.9861	[0.985, 0.987]	7.86×10^{-4}	5	0
vanilla	seq2seq	0	1	[1, 1]	0	5	0
vanilla	seq2seq	0.01	1	[1, 1]	0	5	0
vanilla	seq2seq	0.1	1	[1, 1]	0	5	0
vanilla	seq2seq	1	1	[1, 1]	0	5	0
vanilla	seq2seq	10	1	[1, 1]	0	5	0
vanilla	seq2seq	100	1	[1, 1]	0	5	0
vanilla	seq2seq	1000	1	[1, 1]	0	5	0
vanilla	seq2seq	1e+04	1	[1, 1]	0	5	0

Table 26: `id_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. rank: Wilcoxon signed-rank by seed, Holm.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	0	[0, 0]	–
adaptive	pointer	0.1	0	[0, 0]	–
adaptive	pointer	1	0	[0, 0]	–
adaptive	pointer	10	0	[0, 0]	–
adaptive	pointer	100	0	[0, 0]	–
adaptive	pointer	1000	0	[0, 0]	–
adaptive	pointer	1e+04	0	[0, 0]	–
adaptive	seq2seq	0.01	0	[0, 0]	–
adaptive	seq2seq	0.1	0	[0, 0]	–
adaptive	seq2seq	1	0	[0, 0]	–
adaptive	seq2seq	10	0	[0, 0]	–

Table 26: `id_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. rank: Wilcoxon signed-rank by seed, Holm.

`model_type`	`head`	`fishr_lambda`	difference	95% CI	p
adaptive	seq2seq	100	0	[0, 0]	–
adaptive	seq2seq	1000	0	[0, 0]	–
adaptive	seq2seq	1e+04	0	[0, 0]	–
tropical	pointer	0.01	0	[0, 0]	–
tropical	pointer	0.1	0	[0, 0]	–
tropical	pointer	1	0	[0, 0]	–
tropical	pointer	10	0	[0, 0]	–
tropical	pointer	100	0	[0, 0]	–
tropical	pointer	1000	0	[0, 0]	–
tropical	pointer	1e+04	0	[0, 0]	–
tropical	seq2seq	0.01	0	[0, 0]	–
tropical	seq2seq	0.1	0	[0, 0]	–
tropical	seq2seq	1	0	[0, 0]	–
tropical	seq2seq	10	0	[0, 0]	–
tropical	seq2seq	100	0	[0, 0]	–
tropical	seq2seq	1000	0	[0, 0]	–
tropical	seq2seq	1e+04	0	[0, 0]	–
vanilla	pointer	0.01	0	[0, 0]	–
vanilla	pointer	0.1	0	[0, 0]	–
vanilla	pointer	1	0	[0, 0]	–
vanilla	pointer	10	0	[0, 0]	–
vanilla	pointer	100	0	[0, 0]	–
vanilla	pointer	1000	0	[0, 0]	–
vanilla	pointer	1e+04	0	[0, 0]	–
vanilla	seq2seq	0.01	0	[0, 0]	–
vanilla	seq2seq	0.1	0	[0, 0]	–
vanilla	seq2seq	1	0	[0, 0]	–
vanilla	seq2seq	10	0	[0, 0]	–
vanilla	seq2seq	100	0	[0, 0]	–
vanilla	seq2seq	1000	0	[0, 0]	–
vanilla	seq2seq	1e+04	0	[0, 0]	–

Table 27: `id_f1` against log10 `fishr_lambda`: rank: Spearman rho, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	0	[0, 0]	1.0000	0	1.0000
adaptive	seq2seq	–	[–, –]	–	–	–
tropical	pointer	0	[0, 0]	1.0000	0	1.0000
tropical	seq2seq	–	[–, –]	–	–	–
vanilla	pointer	0	[0, 0]	1.0000	0	1.0000
vanilla	seq2seq	–	[–, –]	–	–	–

Table 28: `id_f1`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	0	[–, –]	–
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0	[–, –]	–

Table 28: `id_f1`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0	[–, –]	–
model_type: tropical - vanilla head=pointer fishr_lambda=1	0	[–, –]	–
model_type: tropical - vanilla head=pointer fishr_lambda=10	0	[–, –]	–
model_type: tropical - vanilla head=pointer fishr_lambda=100	0	[–, –]	–
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0	[–, –]	–
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0	[–, –]	–
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0	[–, –]	–
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0	[–, –]	–
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0	[–, –]	–
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0	[–, –]	–
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0	[–, –]	–
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0	[–, –]	–
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	0	[–, –]	–
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0	[–, –]	–
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0	[–, –]	–
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	0	[–, –]	–
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	0	[–, –]	–
model_type: adaptive - vanilla head=pointer fishr_lambda=1	0	[–, –]	–
model_type: adaptive - vanilla head=pointer fishr_lambda=10	0	[–, –]	–
model_type: adaptive - vanilla head=pointer fishr_lambda=100	0	[–, –]	–
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0	[–, –]	–
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	0	[–, –]	–

Table 28: `id_f1`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	0	[–, –]	–
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	-0.01394	[-0.0149, -0.013]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	-0.01394	[-0.0149, -0.013]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	-0.01394	[-0.0149, -0.013]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	-0.01394	[-0.0149, -0.013]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	-0.01394	[-0.0149, -0.013]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	-0.01394	[-0.0149, -0.013]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	-0.01394	[-0.0149, -0.013]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	-0.01394	[-0.0149, -0.013]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0	-0.01394	[-0.0149, -0.013]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	-0.01394	[-0.0149, -0.013]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	-0.01394	[-0.0149, -0.013]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1	-0.01394	[-0.0149, -0.013]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=10	-0.01394	[-0.0149, -0.013]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=100	-0.01394	[-0.0149, -0.013]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	-0.01394	[-0.0149, -0.013]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	-0.01394	[-0.0149, -0.013]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	-0.01394	[-0.0149, -0.013]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	-0.01394	[-0.0149, -0.013]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	-0.01394	[-0.0149, -0.013]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	-0.01394	[-0.0149, -0.013]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	-0.01394	[-0.0149, -0.013]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	-0.01394	[-0.0149, -0.013]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	-0.01394	[-0.0149, -0.013]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	-0.01394	[-0.0149, -0.013]	1.0000

A.7 id_exact

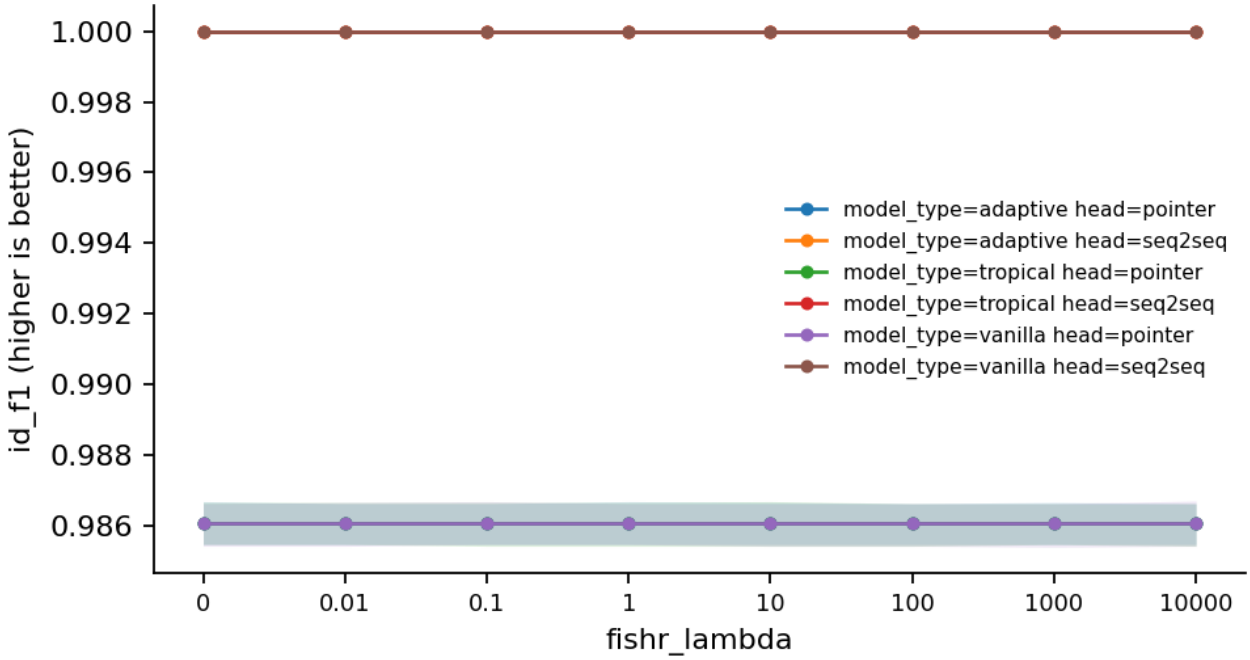


Figure 7: `id_f1` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

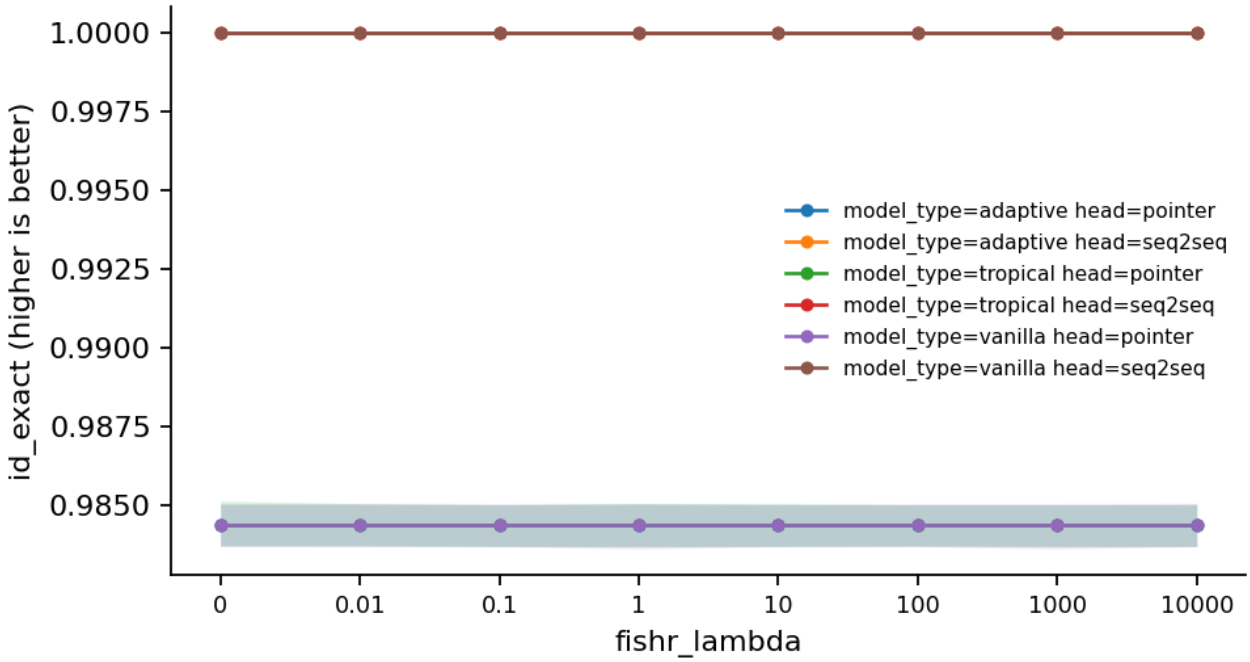


Figure 8: `id_exact` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

Table 29: `id_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.9844	[0.984, 0.985]	8.83×10^{-4}	5	0
adaptive	pointer	0.01	0.9844	[0.984, 0.985]	8.83×10^{-4}	5	0
adaptive	pointer	0.1	0.9844	[0.984, 0.985]	8.83×10^{-4}	5	0
adaptive	pointer	1	0.9844	[0.984, 0.985]	8.83×10^{-4}	5	0
adaptive	pointer	10	0.9844	[0.984, 0.985]	8.83×10^{-4}	5	0
adaptive	pointer	100	0.9844	[0.984, 0.985]	8.83×10^{-4}	5	0
adaptive	pointer	1000	0.9844	[0.984, 0.985]	8.83×10^{-4}	5	0
adaptive	pointer	1e+04	0.9844	[0.984, 0.985]	8.83×10^{-4}	5	0
adaptive	seq2seq	0	1	[1, 1]	0	5	0
adaptive	seq2seq	0.01	1	[1, 1]	0	5	0
adaptive	seq2seq	0.1	1	[1, 1]	0	5	0
adaptive	seq2seq	1	1	[1, 1]	0	5	0
adaptive	seq2seq	10	1	[1, 1]	0	5	0
adaptive	seq2seq	100	1	[1, 1]	0	5	0
adaptive	seq2seq	1000	1	[1, 1]	0	5	0
adaptive	seq2seq	1e+04	1	[1, 1]	0	5	0
tropical	pointer	0	0.9844	[0.984, 0.985]	8.83×10^{-4}	5	0
tropical	pointer	0.01	0.9844	[0.984, 0.985]	8.83×10^{-4}	5	0
tropical	pointer	0.1	0.9844	[0.984, 0.985]	8.83×10^{-4}	5	0
tropical	pointer	1	0.9844	[0.984, 0.985]	8.83×10^{-4}	5	0
tropical	pointer	10	0.9844	[0.984, 0.985]	8.83×10^{-4}	5	0
tropical	pointer	100	0.9844	[0.984, 0.985]	8.83×10^{-4}	5	0
tropical	pointer	1000	0.9844	[0.984, 0.985]	8.83×10^{-4}	5	0
tropical	pointer	1e+04	0.9844	[0.984, 0.985]	8.83×10^{-4}	5	0
tropical	seq2seq	0	1	[1, 1]	0	5	0
tropical	seq2seq	0.01	1	[1, 1]	0	5	0
tropical	seq2seq	0.1	1	[1, 1]	0	5	0
tropical	seq2seq	1	1	[1, 1]	0	5	0
tropical	seq2seq	10	1	[1, 1]	0	5	0
tropical	seq2seq	100	1	[1, 1]	0	5	0
tropical	seq2seq	1000	1	[1, 1]	0	5	0
tropical	seq2seq	1e+04	1	[1, 1]	0	5	0
vanilla	pointer	0	0.9844	[0.984, 0.985]	8.83×10^{-4}	5	0
vanilla	pointer	0.01	0.9844	[0.984, 0.985]	8.83×10^{-4}	5	0
vanilla	pointer	0.1	0.9844	[0.984, 0.985]	8.83×10^{-4}	5	0
vanilla	pointer	1	0.9844	[0.984, 0.985]	8.83×10^{-4}	5	0
vanilla	pointer	10	0.9844	[0.984, 0.985]	8.83×10^{-4}	5	0
vanilla	pointer	100	0.9844	[0.984, 0.985]	8.83×10^{-4}	5	0
vanilla	pointer	1000	0.9844	[0.984, 0.985]	8.83×10^{-4}	5	0
vanilla	pointer	1e+04	0.9844	[0.984, 0.985]	8.83×10^{-4}	5	0
vanilla	seq2seq	0	1	[1, 1]	0	5	0
vanilla	seq2seq	0.01	1	[1, 1]	0	5	0
vanilla	seq2seq	0.1	1	[1, 1]	0	5	0
vanilla	seq2seq	1	1	[1, 1]	0	5	0
vanilla	seq2seq	10	1	[1, 1]	0	5	0
vanilla	seq2seq	100	1	[1, 1]	0	5	0
vanilla	seq2seq	1000	1	[1, 1]	0	5	0
vanilla	seq2seq	1e+04	1	[1, 1]	0	5	0

Table 30: `id_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. rank: Wilcoxon signed-rank by seed, Holm.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	0	[0, 0]	–
adaptive	pointer	0.1	0	[0, 0]	–
adaptive	pointer	1	0	[0, 0]	–
adaptive	pointer	10	0	[0, 0]	–
adaptive	pointer	100	0	[0, 0]	–
adaptive	pointer	1000	0	[0, 0]	–
adaptive	pointer	1e+04	0	[0, 0]	–
adaptive	seq2seq	0.01	0	[0, 0]	–
adaptive	seq2seq	0.1	0	[0, 0]	–
adaptive	seq2seq	1	0	[0, 0]	–
adaptive	seq2seq	10	0	[0, 0]	–
adaptive	seq2seq	100	0	[0, 0]	–
adaptive	seq2seq	1000	0	[0, 0]	–
adaptive	seq2seq	1e+04	0	[0, 0]	–
tropical	pointer	0.01	0	[0, 0]	–
tropical	pointer	0.1	0	[0, 0]	–
tropical	pointer	1	0	[0, 0]	–
tropical	pointer	10	0	[0, 0]	–
tropical	pointer	100	0	[0, 0]	–
tropical	pointer	1000	0	[0, 0]	–
tropical	pointer	1e+04	0	[0, 0]	–
tropical	seq2seq	0.01	0	[0, 0]	–
tropical	seq2seq	0.1	0	[0, 0]	–
tropical	seq2seq	1	0	[0, 0]	–
tropical	seq2seq	10	0	[0, 0]	–
tropical	seq2seq	100	0	[0, 0]	–
tropical	seq2seq	1000	0	[0, 0]	–
tropical	seq2seq	1e+04	0	[0, 0]	–
vanilla	pointer	0.01	0	[0, 0]	–
vanilla	pointer	0.1	0	[0, 0]	–
vanilla	pointer	1	0	[0, 0]	–
vanilla	pointer	10	0	[0, 0]	–
vanilla	pointer	100	0	[0, 0]	–
vanilla	pointer	1000	0	[0, 0]	–
vanilla	pointer	1e+04	0	[0, 0]	–
vanilla	seq2seq	0.01	0	[0, 0]	–
vanilla	seq2seq	0.1	0	[0, 0]	–
vanilla	seq2seq	1	0	[0, 0]	–
vanilla	seq2seq	10	0	[0, 0]	–
vanilla	seq2seq	100	0	[0, 0]	–
vanilla	seq2seq	1000	0	[0, 0]	–
vanilla	seq2seq	1e+04	0	[0, 0]	–

Table 31: `id_exact` against log10 `fishr_lambda`: rank: Spearman rho, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

model_type	head	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	0	[0, 0]	1.0000	0	1.0000
adaptive	seq2seq	–	[–, –]	–	–	–
tropical	pointer	0	[0, 0]	1.0000	0	1.0000
tropical	seq2seq	–	[–, –]	–	–	–
vanilla	pointer	0	[0, 0]	1.0000	0	1.0000
vanilla	seq2seq	–	[–, –]	–	–	–

Table 32: `id_exact`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	0	[–, –]	–
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0	[–, –]	–
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0	[–, –]	–
model_type: tropical - vanilla head=pointer fishr_lambda=1	0	[–, –]	–
model_type: tropical - vanilla head=pointer fishr_lambda=10	0	[–, –]	–
model_type: tropical - vanilla head=pointer fishr_lambda=100	0	[–, –]	–
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0	[–, –]	–
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0	[–, –]	–
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0	[–, –]	–
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0	[–, –]	–
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0	[–, –]	–
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0	[–, –]	–
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0	[–, –]	–
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0	[–, –]	–
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	0	[–, –]	–
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0	[–, –]	–
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0	[–, –]	–
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	0	[–, –]	–
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	0	[–, –]	–
model_type: adaptive - vanilla head=pointer fishr_lambda=1	0	[–, –]	–
model_type: adaptive - vanilla head=pointer fishr_lambda=10	0	[–, –]	–
model_type: adaptive - vanilla head=pointer fishr_lambda=100	0	[–, –]	–
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0	[–, –]	–
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	0	[–, –]	–

Table 32: `id_exact`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	0	[–, –]	–
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	-0.0156	[-0.0167, -0.0145]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	-0.0156	[-0.0167, -0.0145]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	-0.0156	[-0.0167, -0.0145]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	-0.0156	[-0.0167, -0.0145]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	-0.0156	[-0.0167, -0.0145]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	-0.0156	[-0.0167, -0.0145]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	-0.0156	[-0.0167, -0.0145]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	-0.0156	[-0.0167, -0.0145]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0	-0.0156	[-0.0167, -0.0145]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	-0.0156	[-0.0167, -0.0145]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	-0.0156	[-0.0167, -0.0145]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1	-0.0156	[-0.0167, -0.0145]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=10	-0.0156	[-0.0167, -0.0145]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=100	-0.0156	[-0.0167, -0.0145]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	-0.0156	[-0.0167, -0.0145]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	-0.0156	[-0.0167, -0.0145]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	-0.0156	[-0.0167, -0.0145]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	-0.0156	[-0.0167, -0.0145]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	-0.0156	[-0.0167, -0.0145]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	-0.0156	[-0.0167, -0.0145]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	-0.0156	[-0.0167, -0.0145]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	-0.0156	[-0.0167, -0.0145]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	-0.0156	[-0.0167, -0.0145]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	-0.0156	[-0.0167, -0.0145]	1.0000

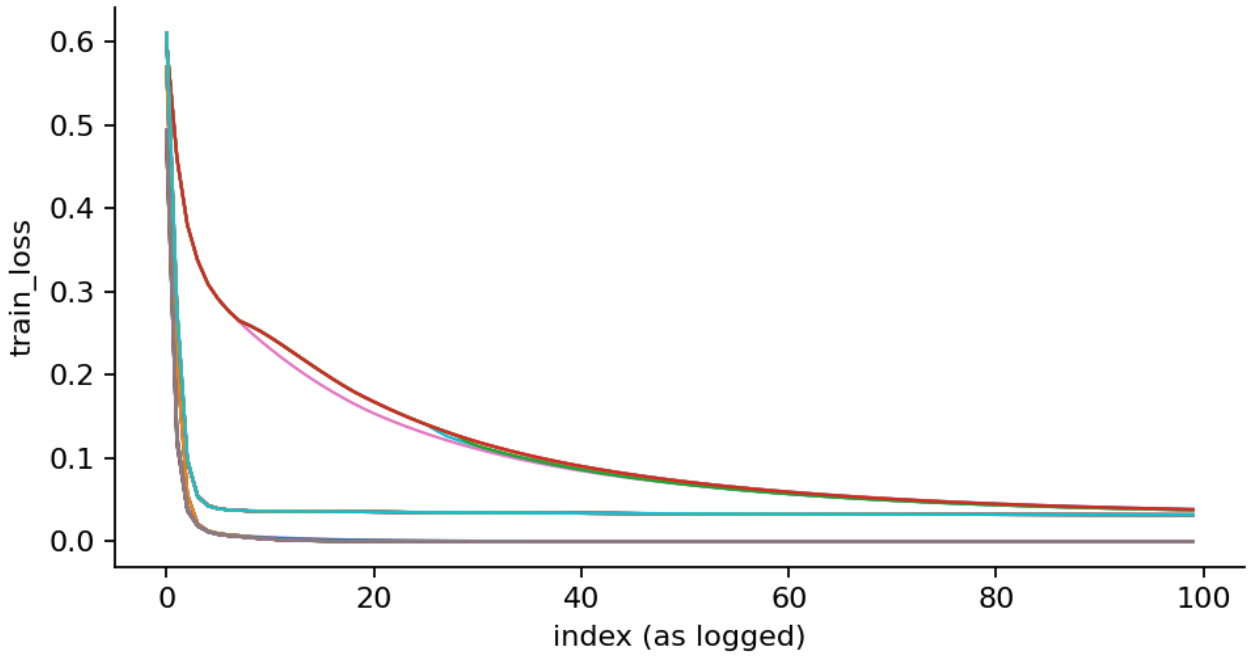


Figure 9: `train_loss`, the mean over seeds per cell.

B Curves

C Configuration

The base configuration every run shares (frozen hash `ff2962d32940`); factor levels are in Table 1.

```
task:
  name: MaxSubsetSumDataset
  value_range:
  - -5
  - 5
  target_range: null
  weight_range: null
  adversarial_range:
  - 10
  - 30
  noise_prob: 0.0
data:
  train_lengths:
  - 4
  - 5
  - 6
  - 7
  - 8
  ood_lengths:
  - 12
  - 16
  - 64
  ood_value_range: null
  ood_noise_range: null
  ood_noise_prob: 0.5
  n_per_length: 20000
  n_eval_per_length: 2000
  curve_eval_limit: 256
model:
  d_model: 64
```

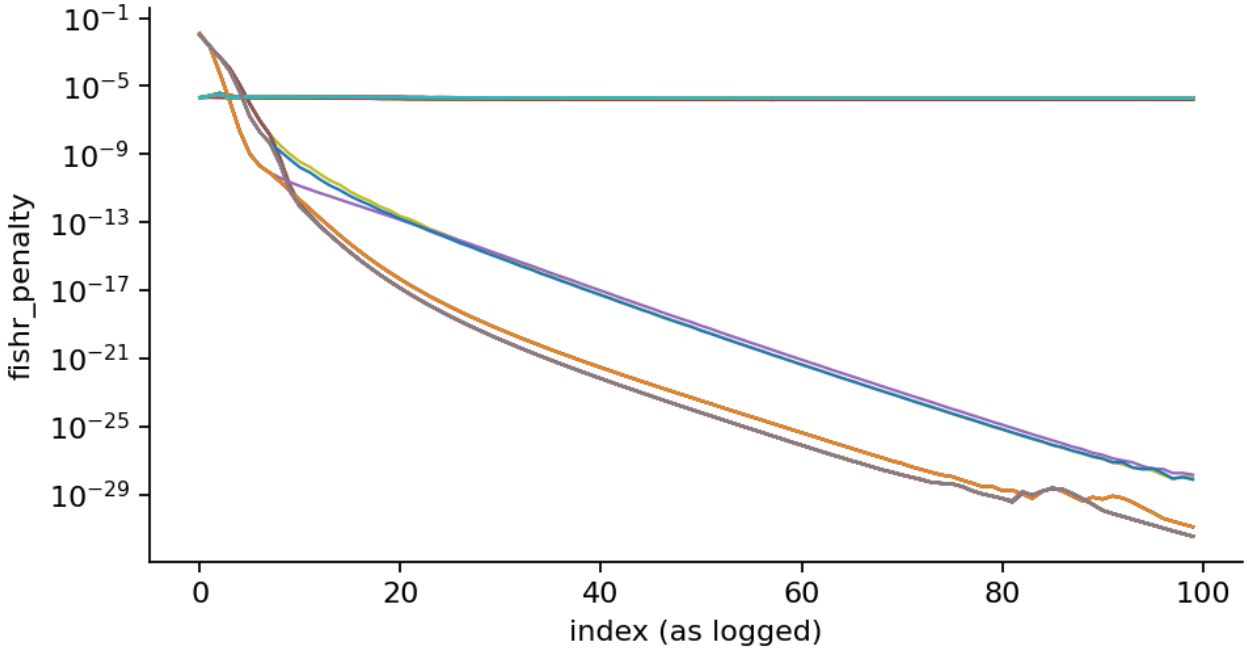


Figure 10: `fishr_penalty`, the mean over seeds per cell.

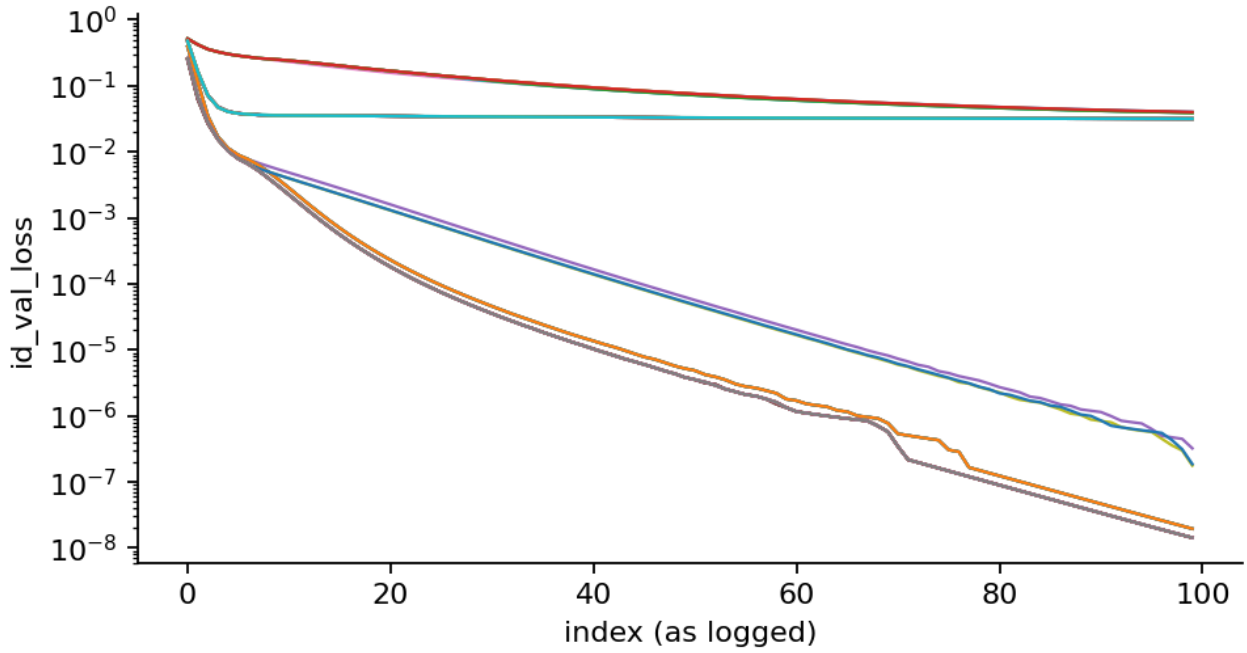


Figure 11: `id_val_loss`, the mean over seeds per cell.

```
  n_heads: 2
  num_layers: 1
  dropout: 0.0
  tropical_impl: legacy
optim:
  lr: 0.0001
  epochs: 100
  batch_size: 100
  eval_batch_size: 256
fishr:
  ema: 0.95
  anneal_iters: 1500
  reset_optimizer: true
```

Store: `campaigns/tropical/MaxSubsetSum/results`.